Type Test Report

Date of issue: 1.9.2015

Serial No.:

Customer:

Customer ref.:

Type: M3AA 250SMA 4
Product Code: 3GAA252210-ADG

Rating:

	V		Hz	kW	r/min	A	cos φ	Duty
3~Motor	690	Y	50	55,0	1480	57,3	0,85	S1
Insul.cl.F	400	D	50	55,0	1480	98,9	0,85	S1
IP55	415	D	50	55,0	1481	96,3	0,84	S1
	440	D	60	55,0	1781	89,7	0,85	S1
	460	D	60	55,0	1782	86,3	0,84	S1

Eff class IE2

50Hz : IE2 - 94,4(100%) - 94,9(75%) - 94,6(50%)
60Hz : IE2 - 94,5(100%)

Resistance
Line — Ambient: 25,6 ºC

U_1 - V_1	0,06718	Ω
U_1 - W_1	0,06603	Ω
V_1 - W_1	0,06722	Ω

Insulation resistance at 26 ºC
R > 2000 Mohm 1000 V

Overload

Current	150 %	120s
Torque	160 %	15s
Speed	120 %	120s

High-voltage test winding 2400 V 60 s

Test	Torque [Nm]	Line U[V]		f[Hz]	Input I[A]	P1 [kW]	Output P2 [kW]	n[r/min]	cos φ	η [%]
No load test		400,8	D	50	35,6	1,04		1500	0,04	
Locked rotor test		71,3	D	50	92,5	4,30		0	0,38	
Thermal test (100% load)	354,9	400	D	50	99,5	58,25	55,00	1480	0,84	94,42
Partial load points:										
~75% load	268,5	400	D	50	78,5	44,04	41,79	1486	0,81	94,85
~50% load	178,9	400	D	50	58,8	29,52	27,94	1491	0,72	94,66
~25% load	90,2	400	D	50	43,4	15,30	14,13	1496	0,51	92,34

Temperature rise at rated load.

	[°C]	[K]	Method
Stator winding :		63,0	1
Frame :		24,0	2
Bearing D-end :		41,9	2
Ambient Temperature :	26		2

Measurement method
1 Resistance
2 Thermometer
3 Thermocouples

These tests have been carried out on motor no. 08 729332 10001, on date 2009-06-03
which is identical in electrical design with the above.

Manufactured and tested in accordance with rules of IEC 60034-1 and IEC 60034-2-1.
PLL determined from residual loss.

On behalf of customer

On behalf of manufacturer Date of test

Tested by ABB AB, LV Motors, 721 70 Västerås, Sweden

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Computer print-out valid without signature.

ZEV tripping curves

3-phase tripping curve

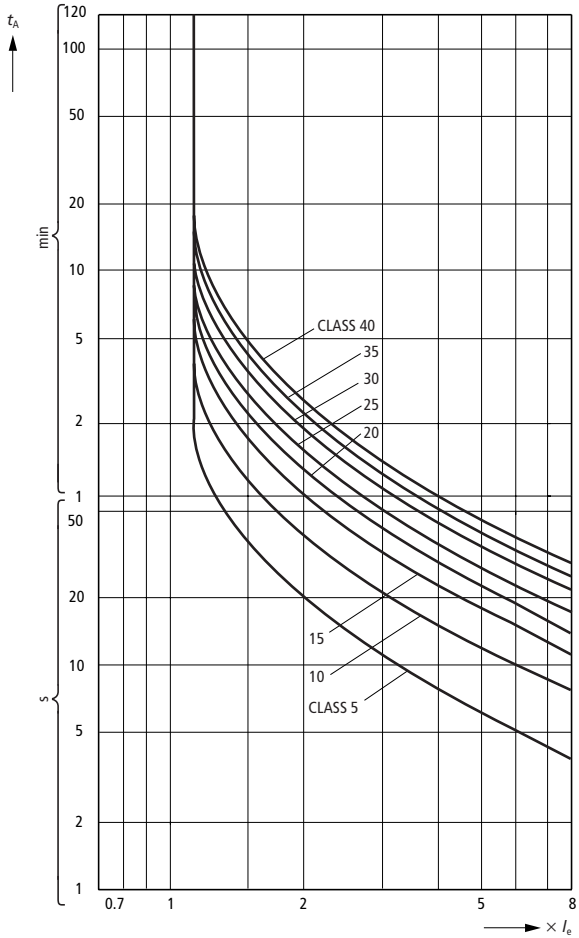


Figure 39: ZEV tripping curve, 3-phase

Table 8: Assignment of the tripping delay to the tripping classes

CLASS	t_A [s]						
	3	**4**	**5**	**6**	**7.2**	**8**	**10**
40	90.5	63.6	49.1	40.0	32.7	29.2	23.0
35	79.2	55.7	43.0	35.0	28.6	25.5	20.1
30	67.9	47.7	36.8	30.0	24.5	21.9	17.2
25	56.6	39.8	30.7	25.0	20.5	18.2	14.4
20	45.3	31.8	24.6	20.0	16.4	14.6	11.5
15	34.0	23.9	18.4	15.0	12.3	10.9	8.6
10	22.6	15.9	12.3	10.0	8.2	7.3	5.7
5	11.3	8.0	6.1	5.0	4.1	3.6	2.9

→ In a 3-phase symmetrical tripping system, the deviation of the tripping delay t_A as of three times the tripping current t_A is ±20 %.

Machine Reactance aggregate calibration | N = 41,877

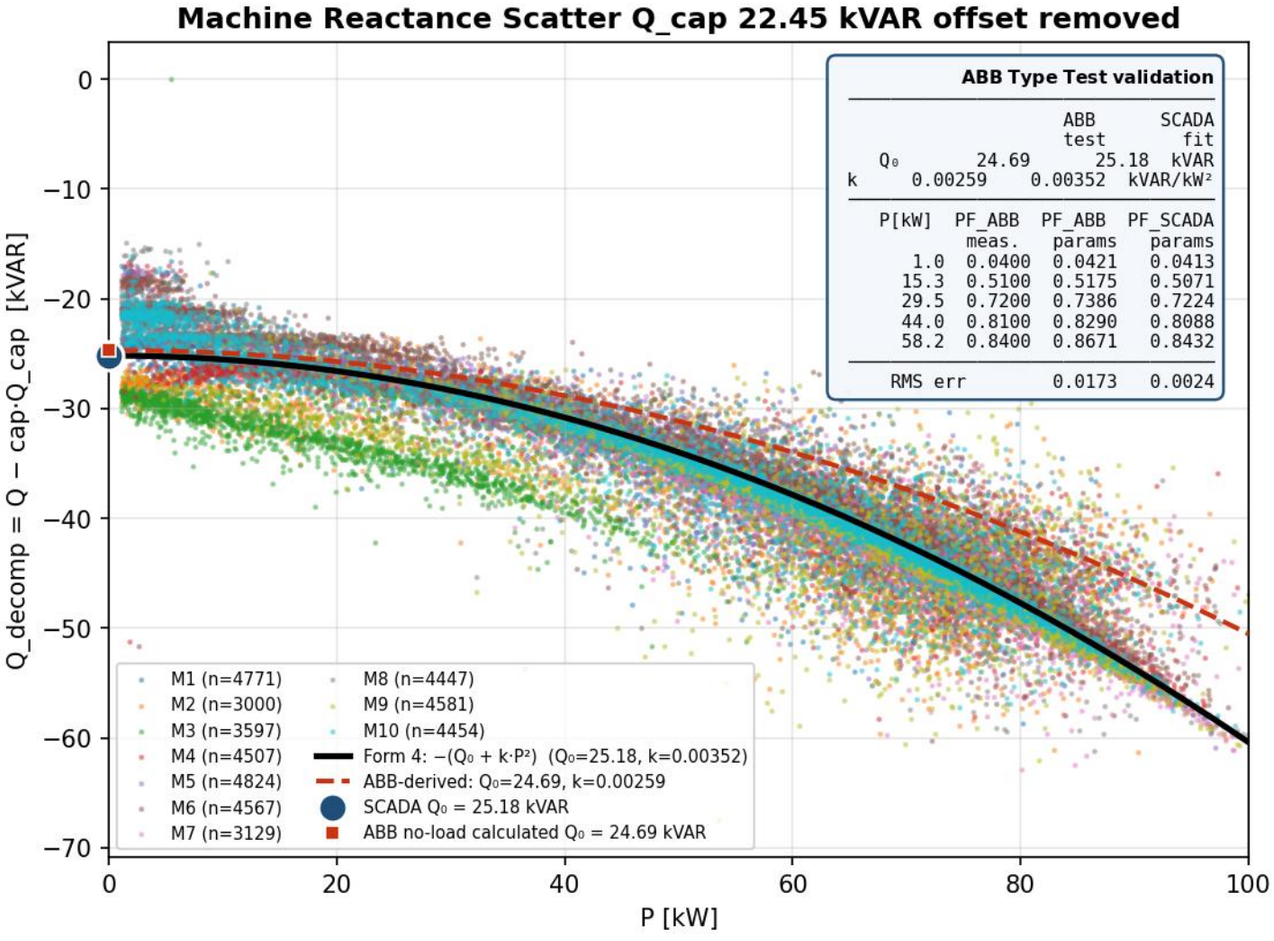
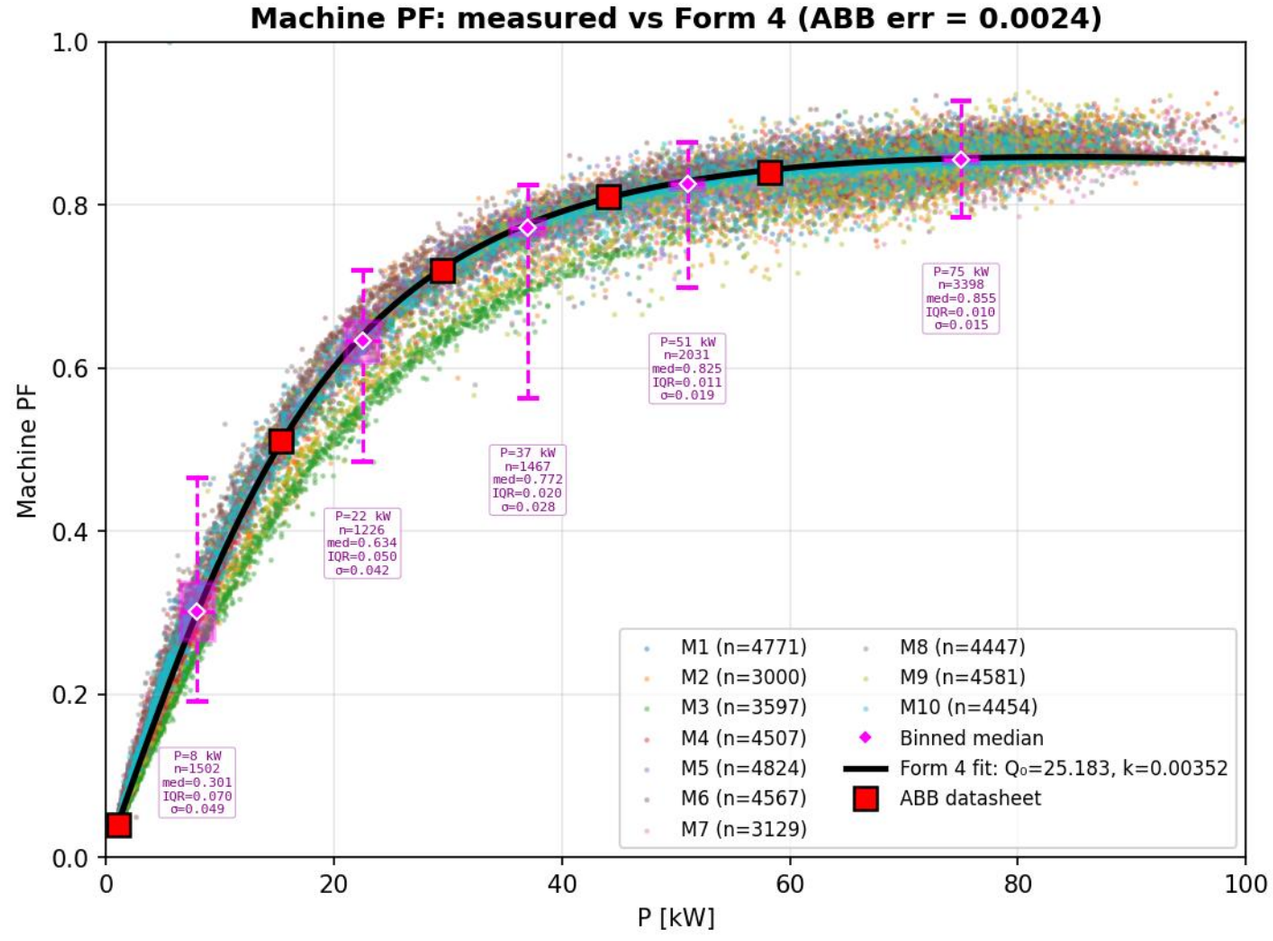


ABB Type Test validation

	ABB test	SCADA fit	
Q_0	24.69	25.18	kVAR
k	0.00259	0.00352	kVAR/kW²

P[kW]	PF_ABB meas.	PF_ABB params	PF_SCADA params
1.0	0.0400	0.0421	0.0413
15.3	0.5100	0.5175	0.5071
29.5	0.7200	0.7386	0.7224
44.0	0.8100	0.8290	0.8088
58.2	0.8400	0.8671	0.8432
RMS err		0.0173	0.0024



ZEV Motor-Protective Relay — Tripping Curves

$$t_A = \frac{CLASS}{5} \times 23.3 \times \ln\left(\frac{x}{x-1.15}\right) \quad \textbf{where } x = I/I_e$$

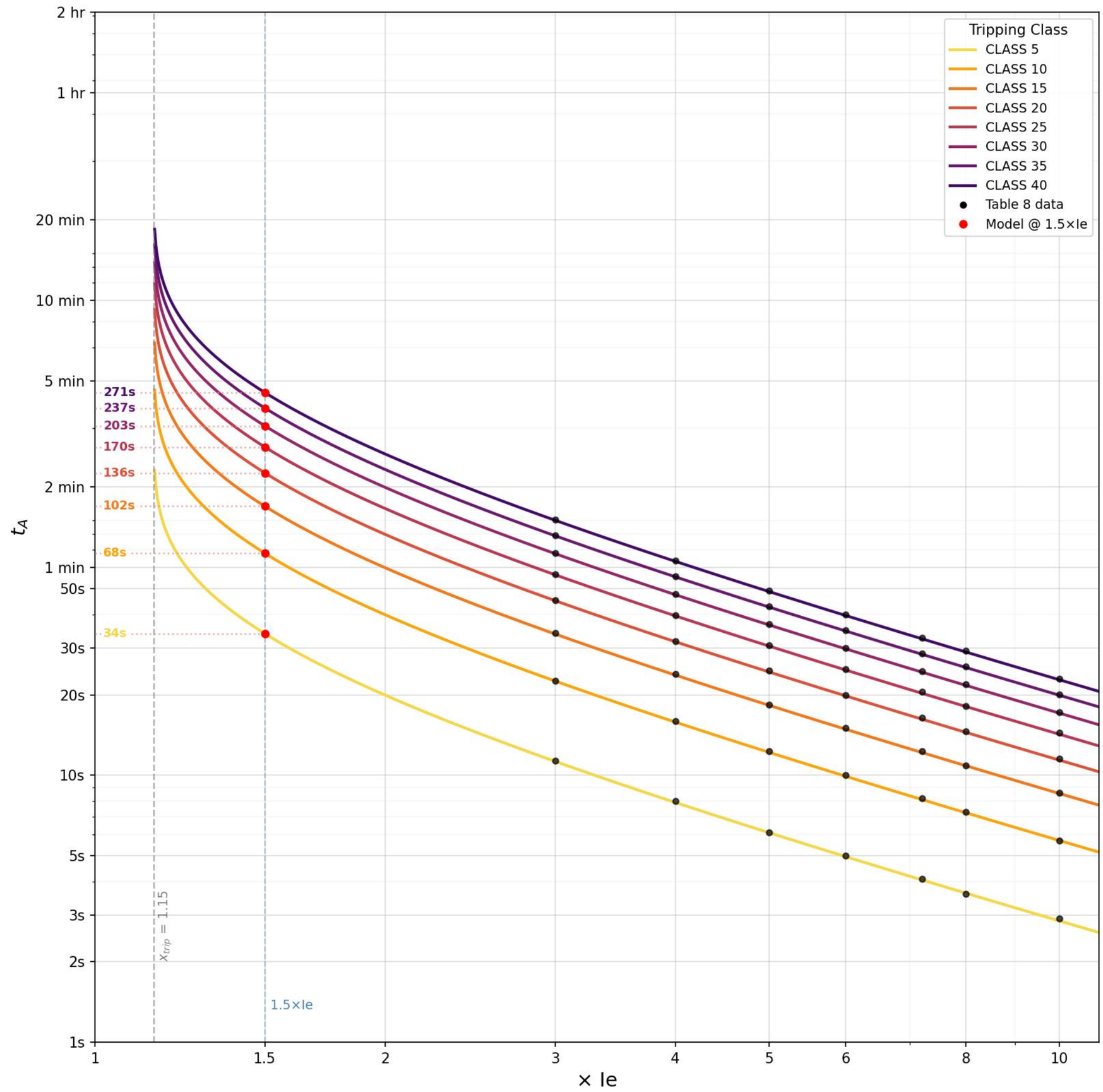


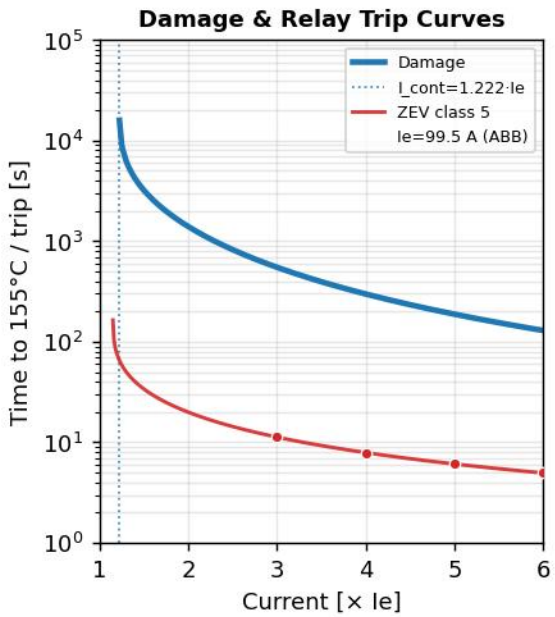
Table 8: Assignment of tripping delay tA [s] to tripping classes

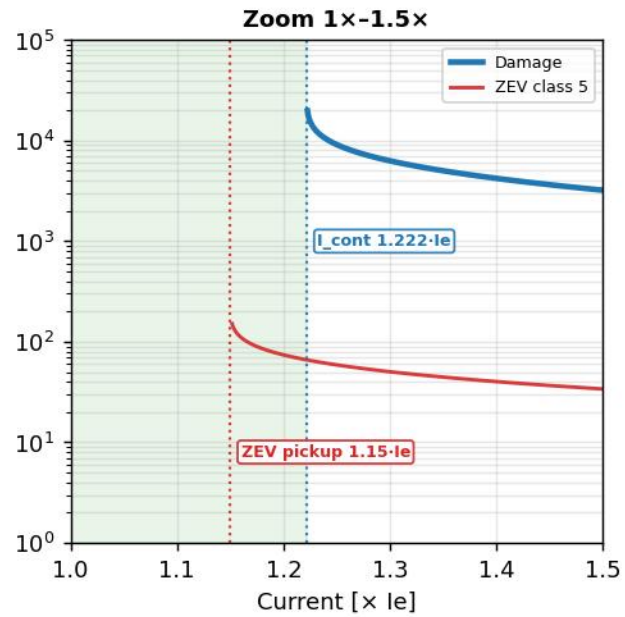
tA [s] at × Ie multiples

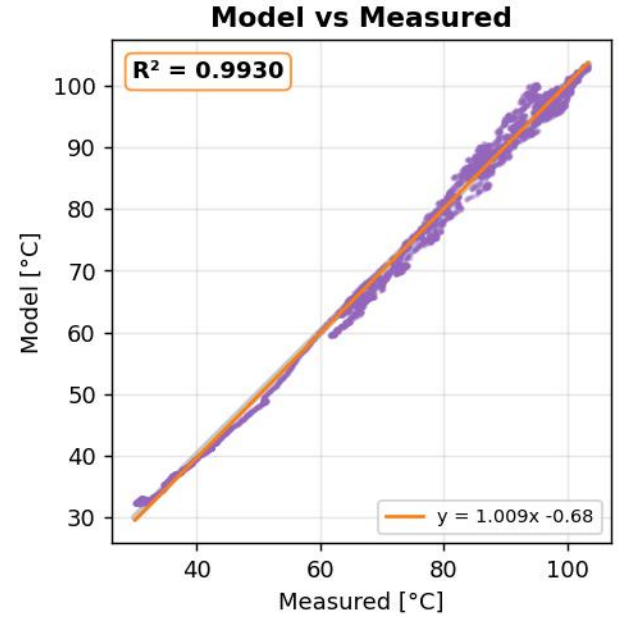
CLASS	3	4	5	6	7.2	8	10
40	90.5	63.6	49.1	40.0	32.7	29.2	23.0
35	79.2	55.7	43.0	35.0	28.6	25.5	20.1
30	67.9	47.7	36.8	30.0	24.5	21.9	17.2
25	56.6	39.8	30.7	25.0	20.5	18.2	14.4
20	45.3	31.8	24.6	20.0	16.4	14.6	11.5
15	34.0	23.9	18.4	15.0	12.3	10.9	8.6
10	22.6	15.9	12.3	10.0	8.2	7.3	5.7
5	11.3	8.0	6.1	5.0	4.1	3.6	2.9

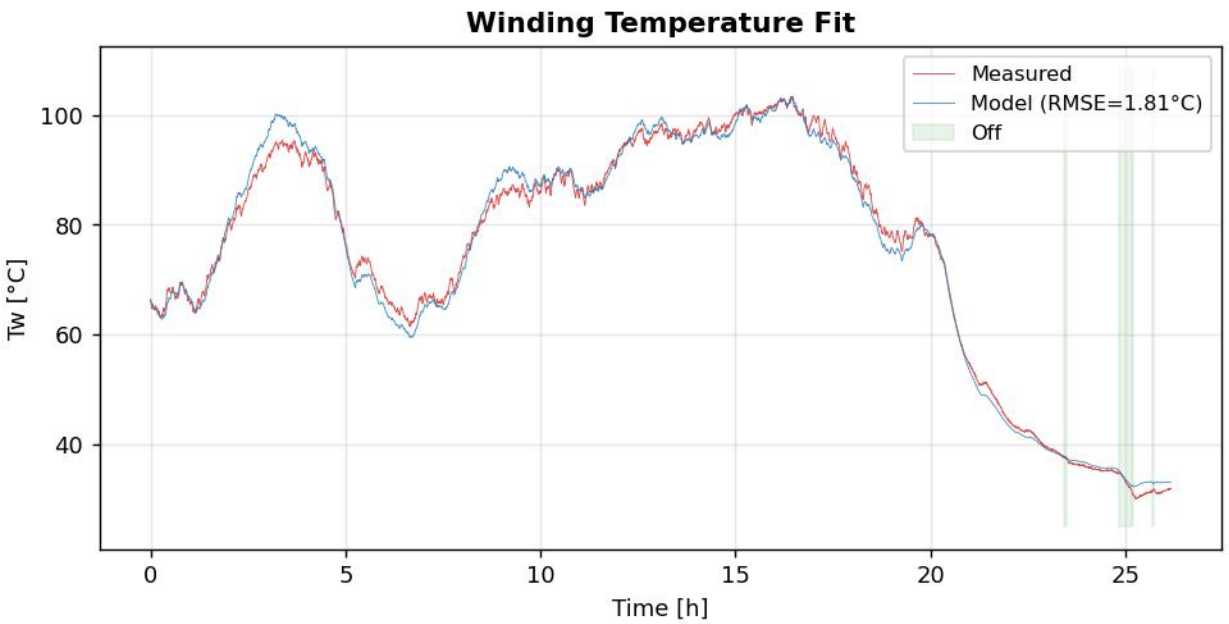
In a 3-phase symmetrical tripping system, the deviation of tA as of 3× the tripping current is ±20%.

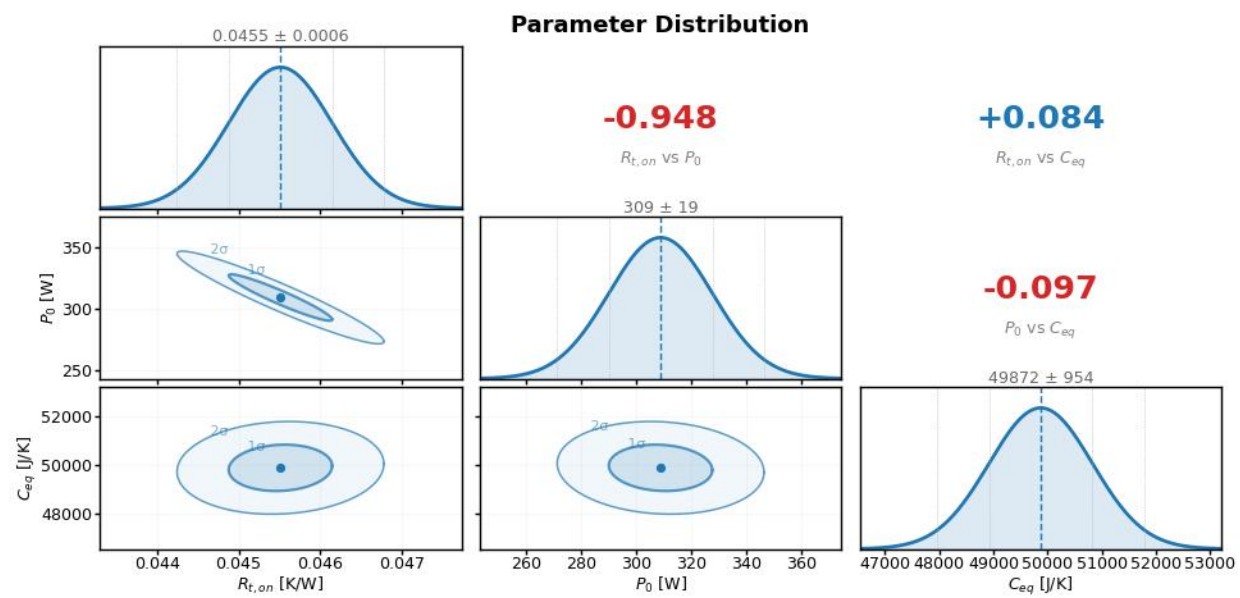
T1 — Machine Study

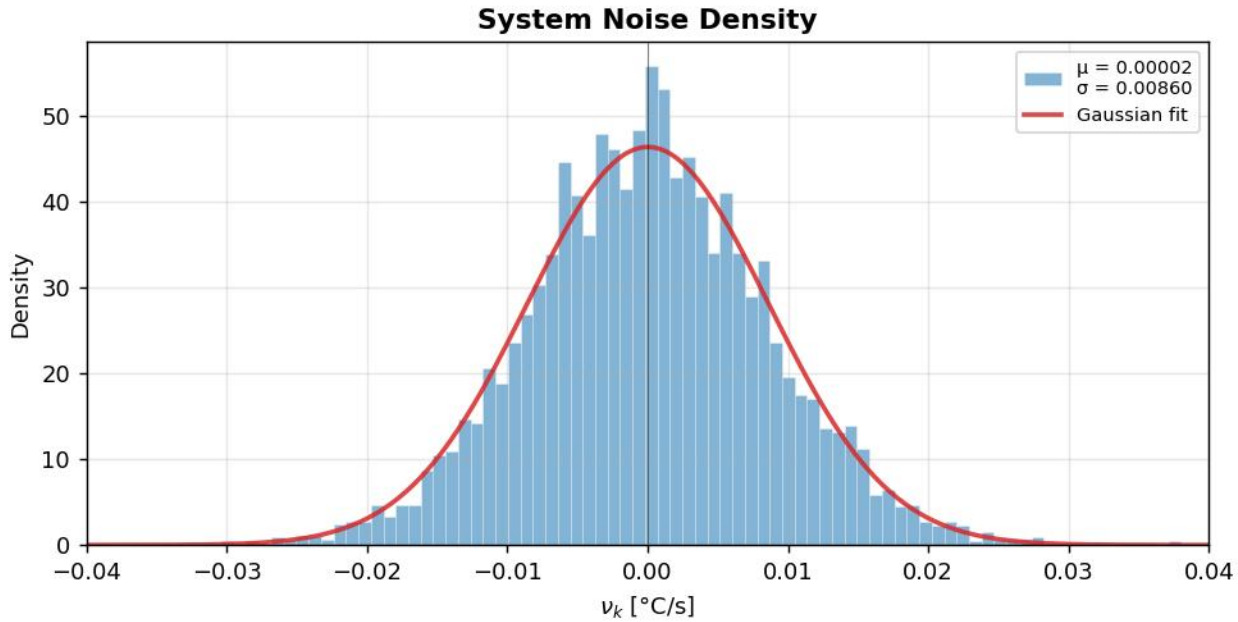












Model Parameters

Fitted parameters

$R_{t,on} = 0.04551 \pm 0.00063 \;\; [°C/W]$

$R_{t,off} = 0.15309 \pm 0.05733 \;\; [°C/W]$

$P_0 = 308.9 \pm 18.8 \;\; [W]$

$C_{eq} = 49872 \pm 954 \;\; [J/°C]$

Derived quantities

$\tau_{on} = R_{t,on} \times C_{eq} = 37.8 \pm 0.9$ min

$I_{cont} = 1.222 \pm 0.005 \;\; \times I_e = 121.6 \pm 0.5$ A

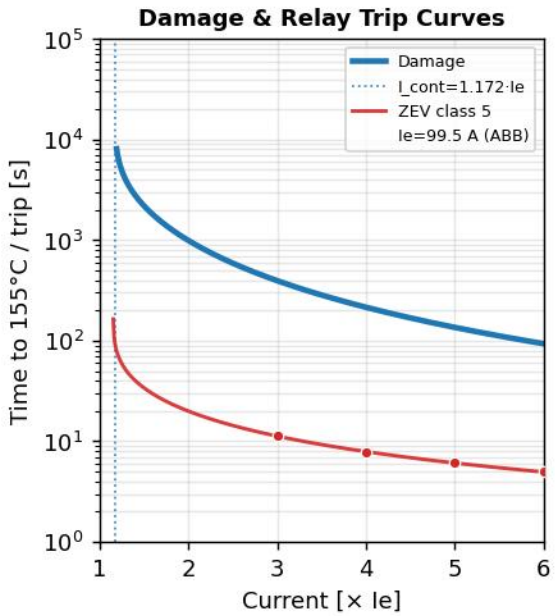
Fit quality

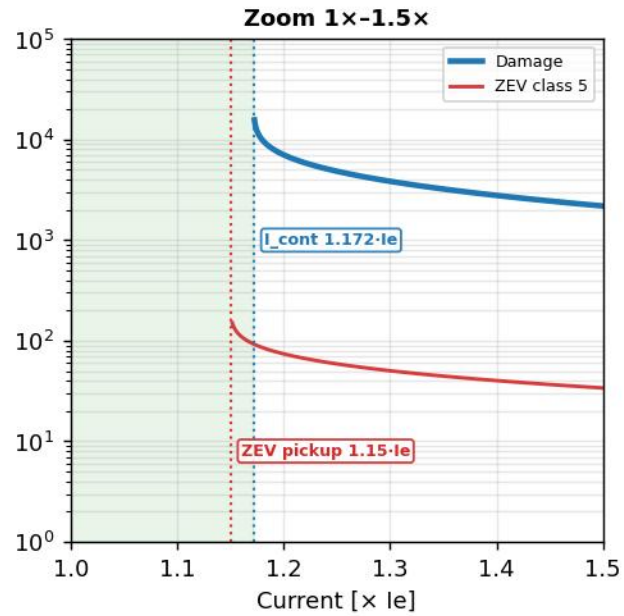
$RMSE = 1.814°C \qquad R^2 = 0.9927 \qquad \sigma(T_w) = 21.28°C$

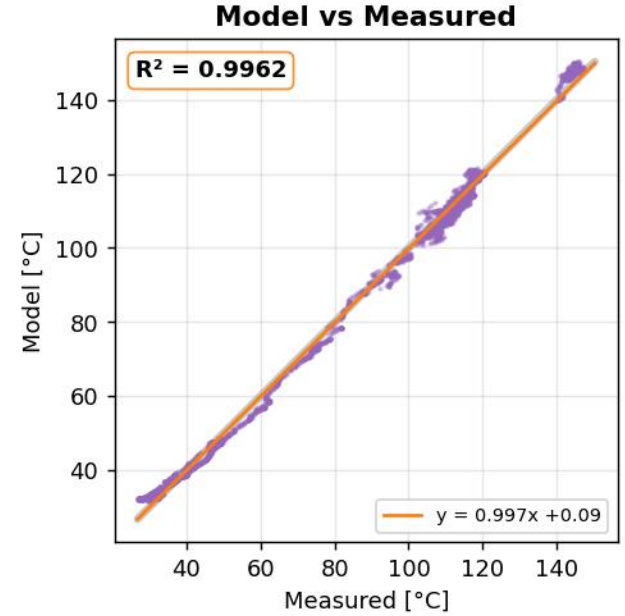
$\sigma_w = 0.00860 \;\; \pm \sim 1\% \;\; [°C/s] \qquad corr(\nu, I) = +0.038$

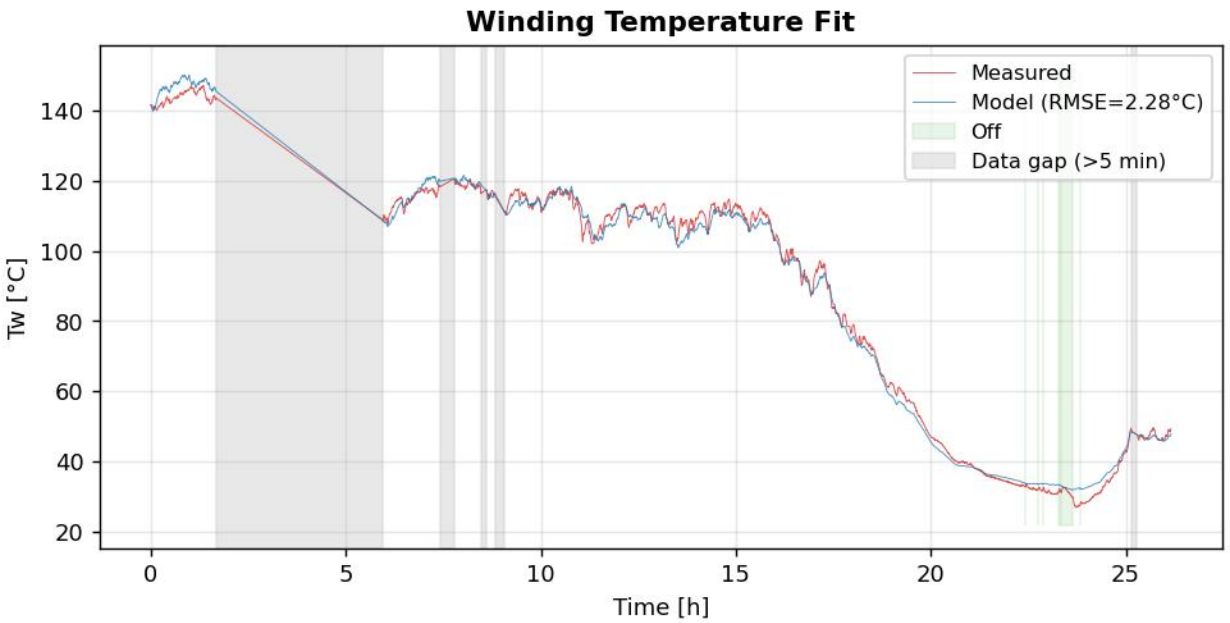
$N_{eff} = 5023 \qquad N = 5023$

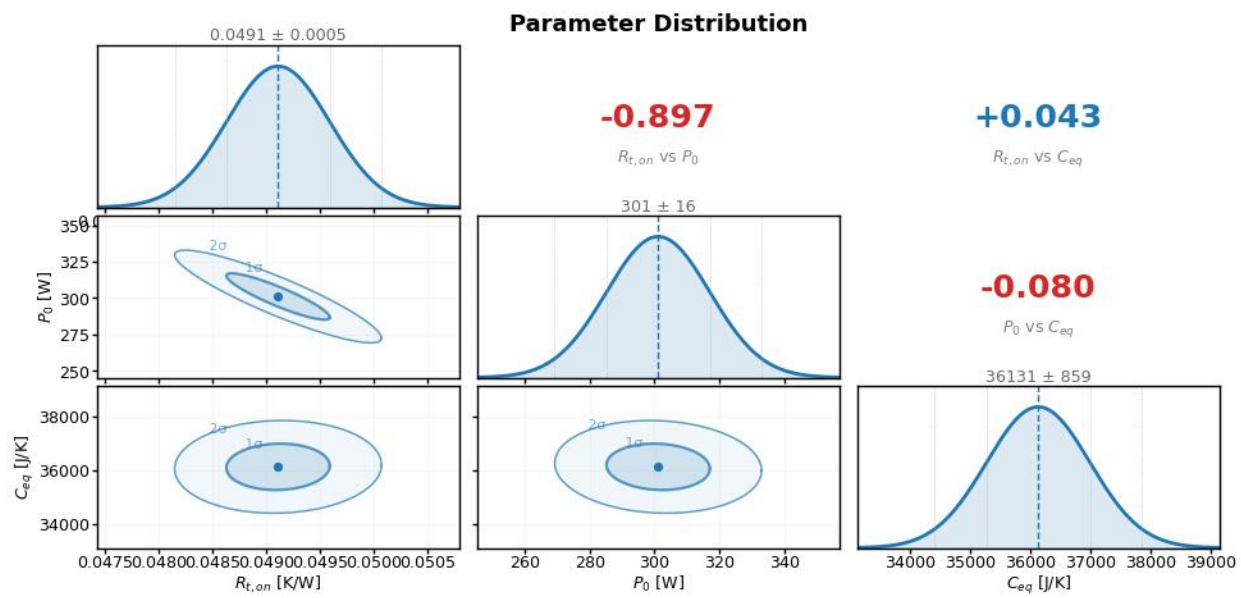
T2 — Machine Study

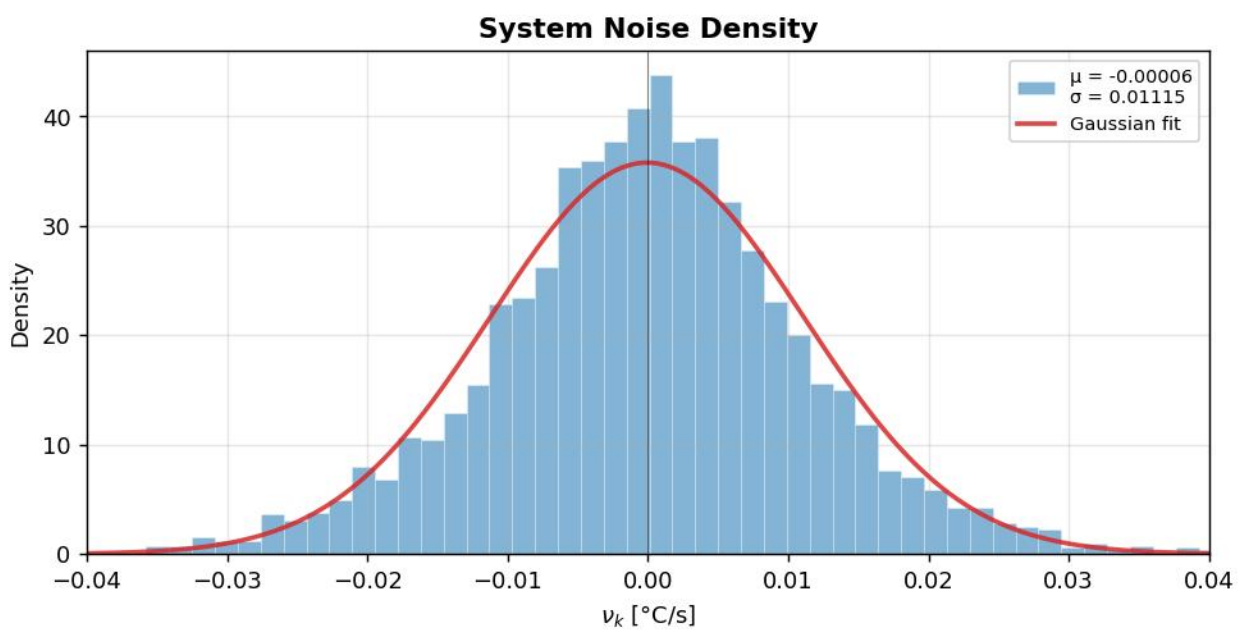












Model Parameters

Fitted parameters

$R_{t,on} = 0.04911 \pm 0.00048 \;\; [°C/W]$

$R_{t,off} = 0.50000 \pm 0.72388 \;\; [°C/W]$

$P_0 = 301.1 \pm 15.9 \;\; [W]$

$C_{eq} = 36131 \pm 859 \;\; [J/°C]$

Derived quantities

$\tau_{on} = R_{t,on} \times C_{eq} = 29.6 \pm 0.8$ min

$I_{cont} = 1.172 \pm 0.003 \;\; \times I_e = 116.6 \pm 0.3$ A

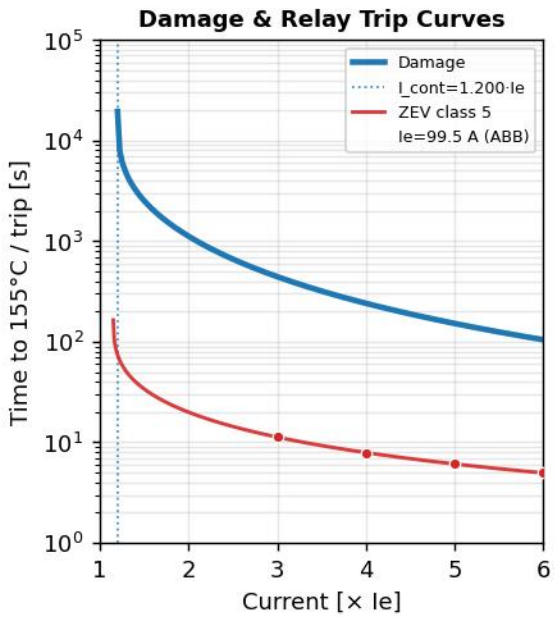
Fit quality

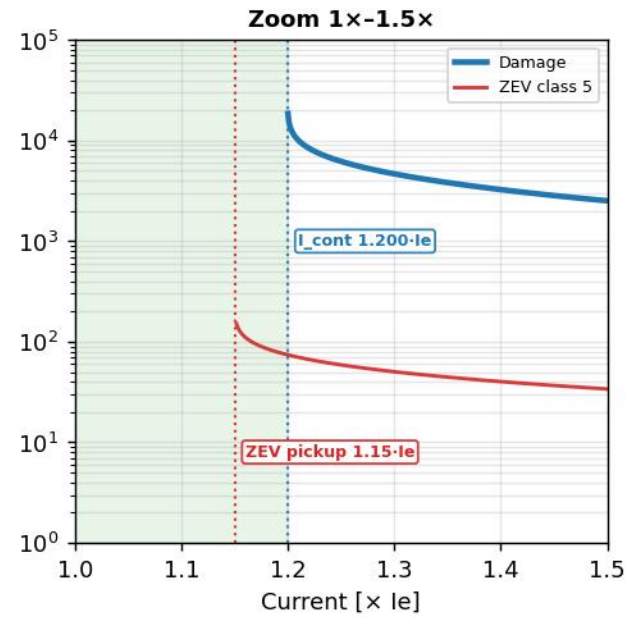
$RMSE = 2.284°C \qquad R^2 = 0.9962 \qquad \sigma(T_w) = 36.91°C$

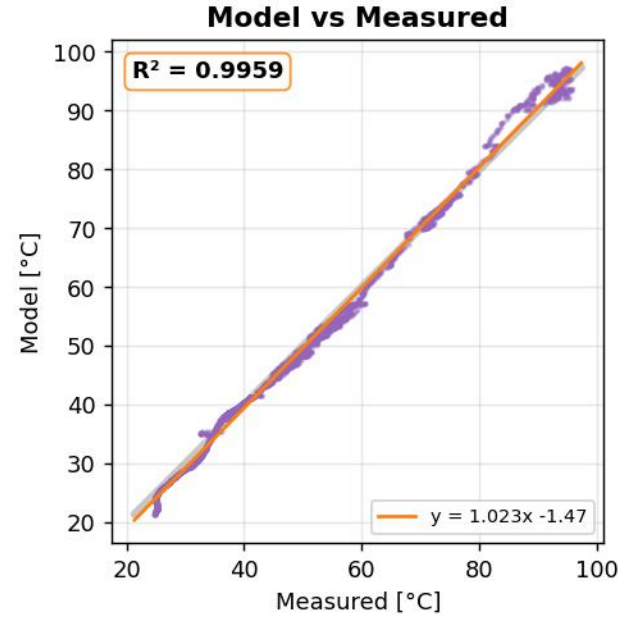
$\sigma_w = 0.01115 \;\; \pm \sim 1\% \;\; [°C/s] \qquad corr(\nu, I) = +0.050$

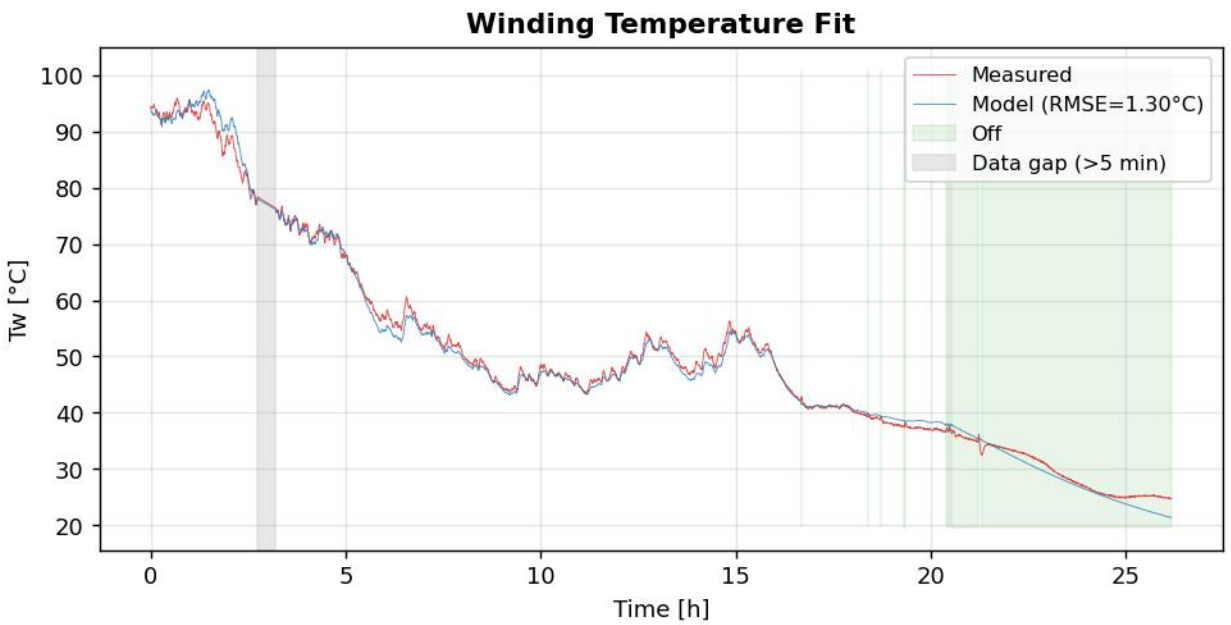
$N_{eff} = 3223 \qquad N = 3228$

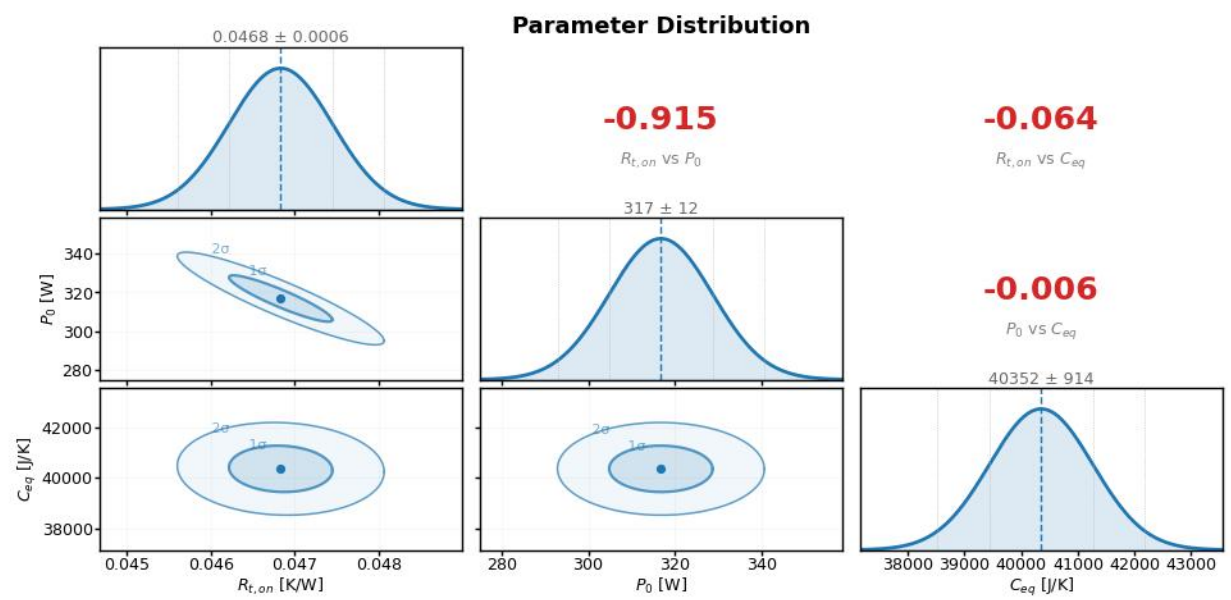
T3 — Machine Study

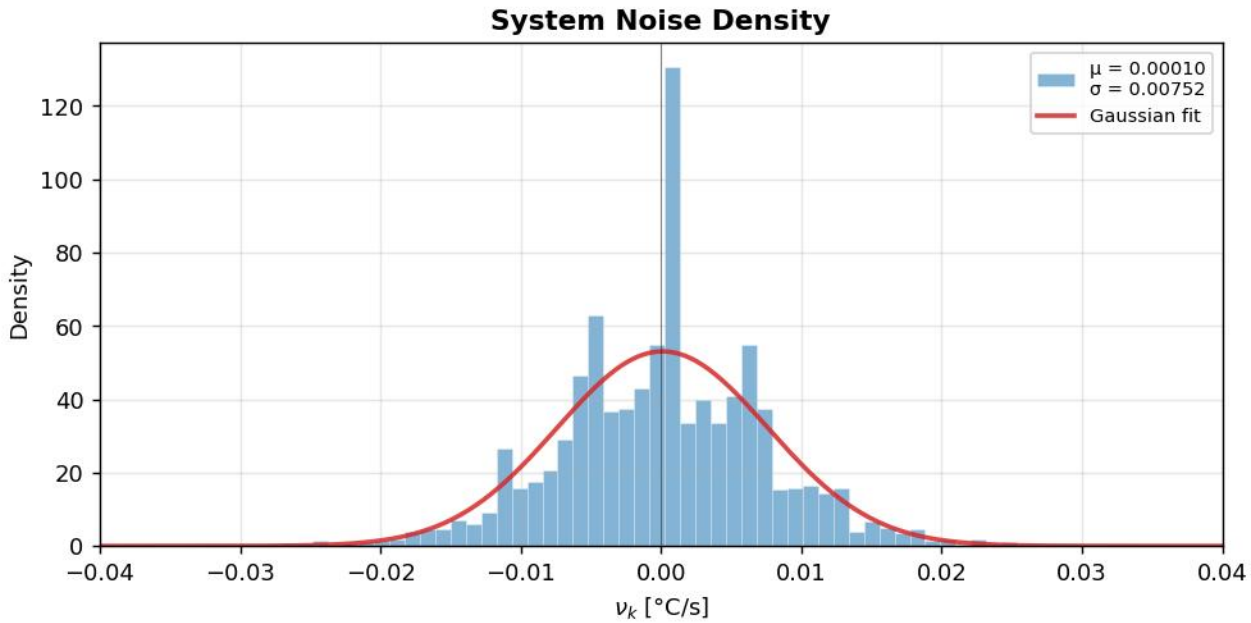












Model Parameters

Fitted parameters

$R_{t,on} = 0.04683 \pm 0.00062 \;\; [°C/W]$

$R_{t,off} = 0.50000 \pm 0.12711 \;\; [°C/W]$

$P_0 = 316.8 \pm 11.9 \;\; [W]$

$C_{eq} = 40352 \pm 914 \;\; [J/°C]$

Derived quantities

$\tau_{on} = R_{t,on} \times C_{eq} = 31.5 \pm 0.8$ min

$I_{cont} = 1.200 \pm 0.006 \;\; \times I_e = 119.4 \pm 0.6$ A

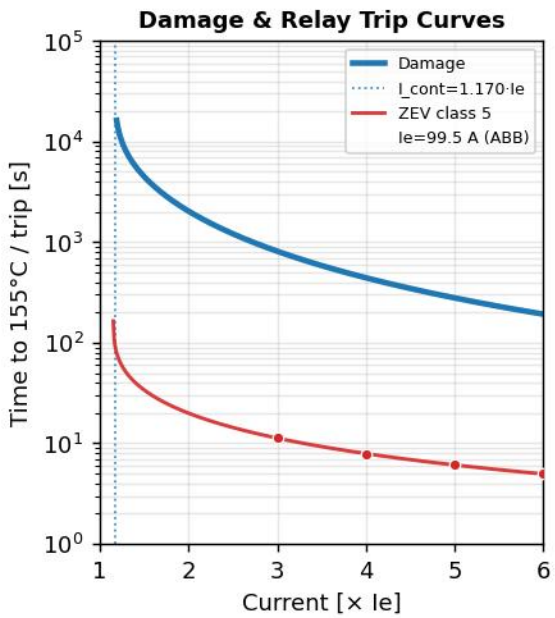
Fit quality

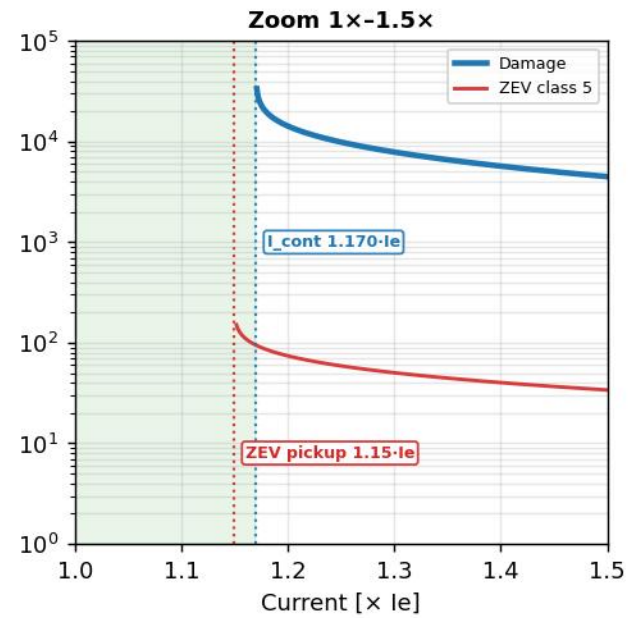
$RMSE = 1.302°C \qquad R^2 = 0.9949 \qquad \sigma(T_w) = 18.16°C$

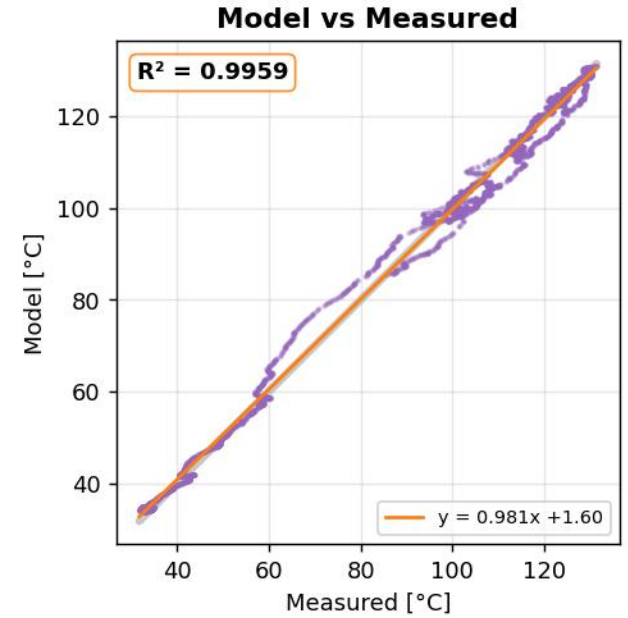
$\sigma_w = 0.00752 \;\; \pm \sim 1\% \;\; [°C/s] \qquad corr(\nu, I) = +0.042$

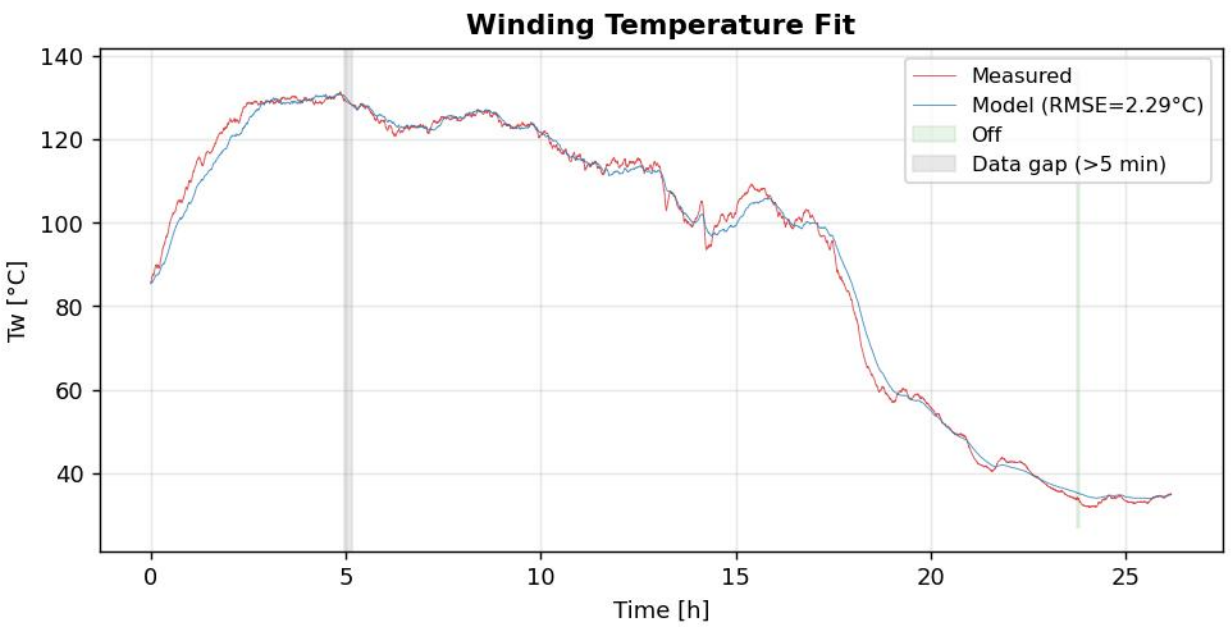
$N_{eff} = 4857 \qquad N = 4858$

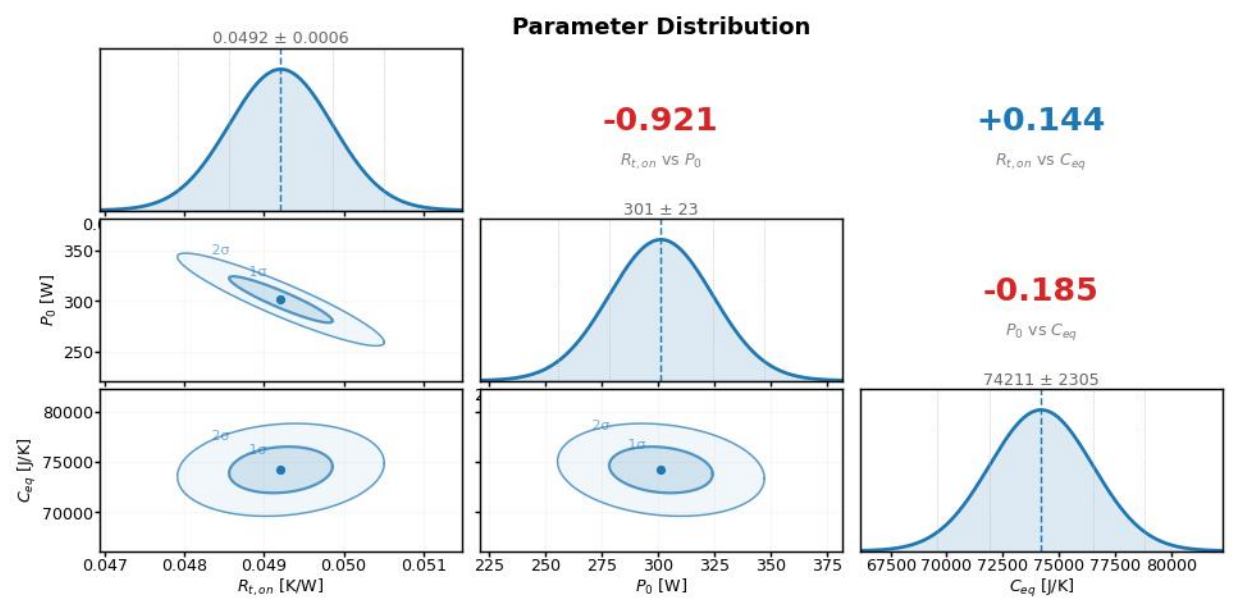
T4 — Machine Study

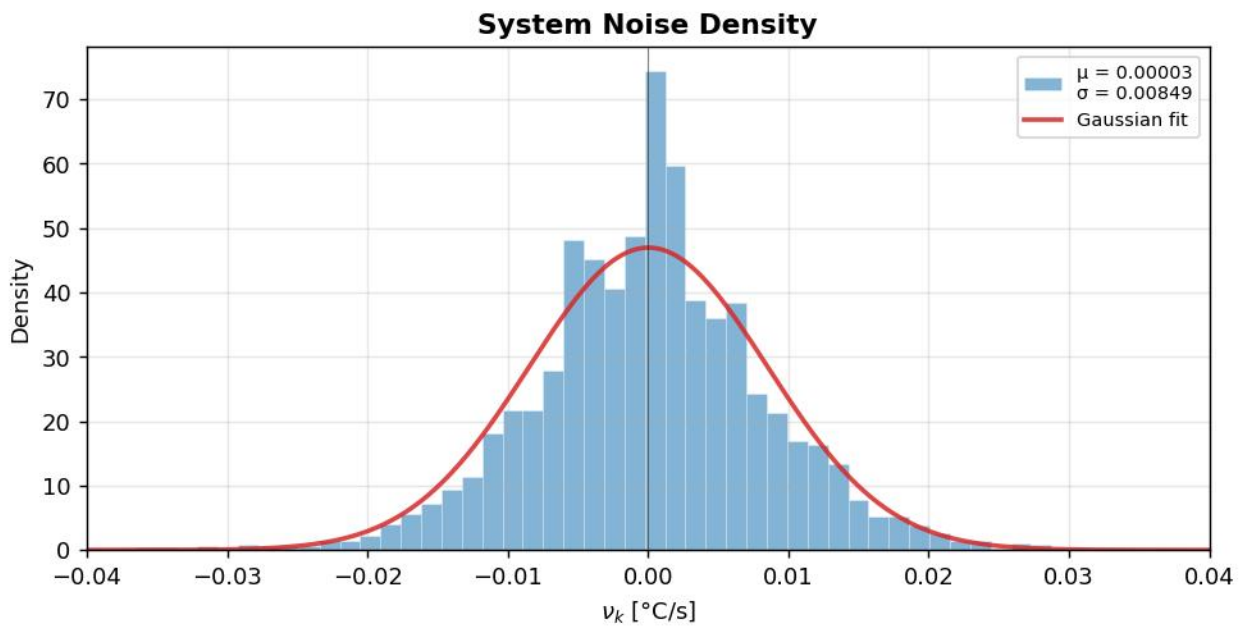












Model Parameters

Fitted parameters

$R_{t,on} = 0.04921 \pm 0.00065 \;\; [°C/W]$

$R_{t,off} = 0.50000 \pm 2.39536 \;\; [°C/W]$

$P_0 = 301.4 \pm 23.0 \;\; [W]$

$C_{eq} = 74211 \pm 2305 \;\; [J/°C]$

Derived quantities

$\tau_{on} = R_{t,on} \times C_{eq} = 60.9 \pm 2.2$ min

$I_{cont} = 1.170 \pm 0.004 \;\times I_e = 116.5 \pm 0.4$ A

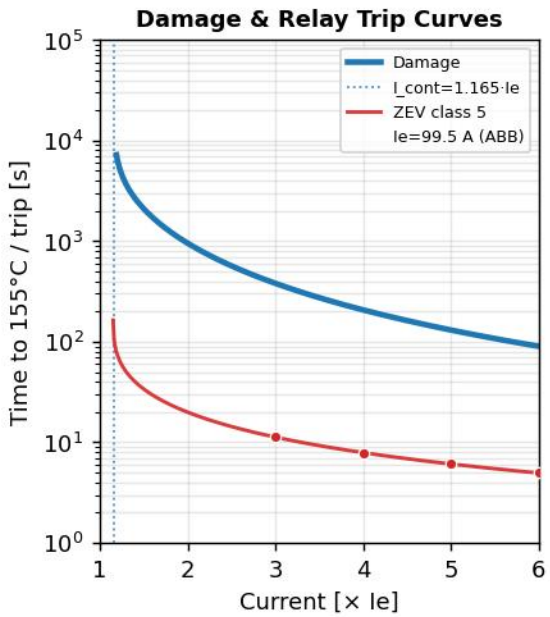
Fit quality

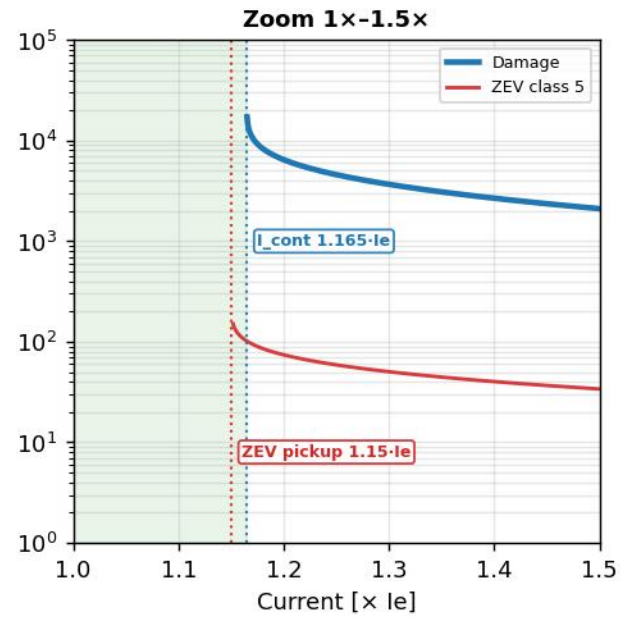
$RMSE = 2.294°C \qquad R^2 = 0.9957 \qquad \sigma(T_w) = 34.87°C$

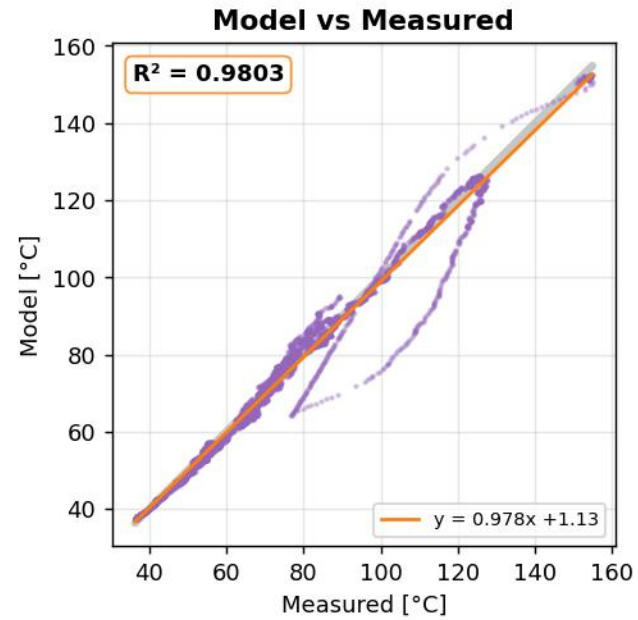
$\sigma_w = 0.00849 \;\pm\; \sim 1\% \;\; [°C/s] \qquad corr(\nu, I) = +0.028$

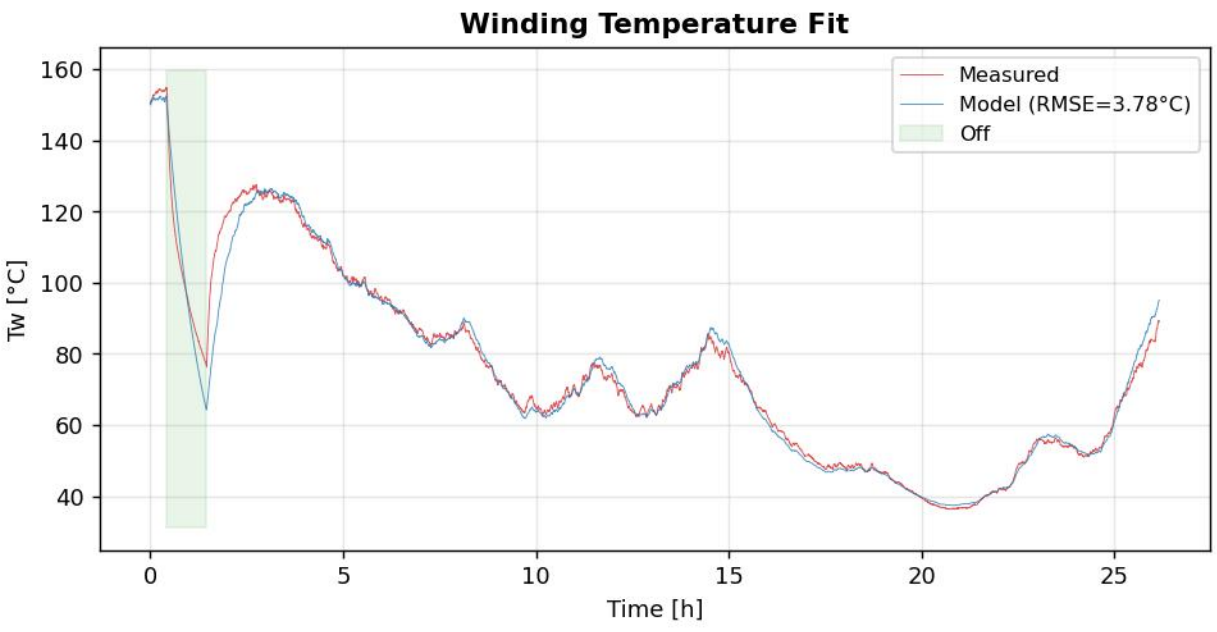
$N_{eff} = 4591 \qquad N = 4592$

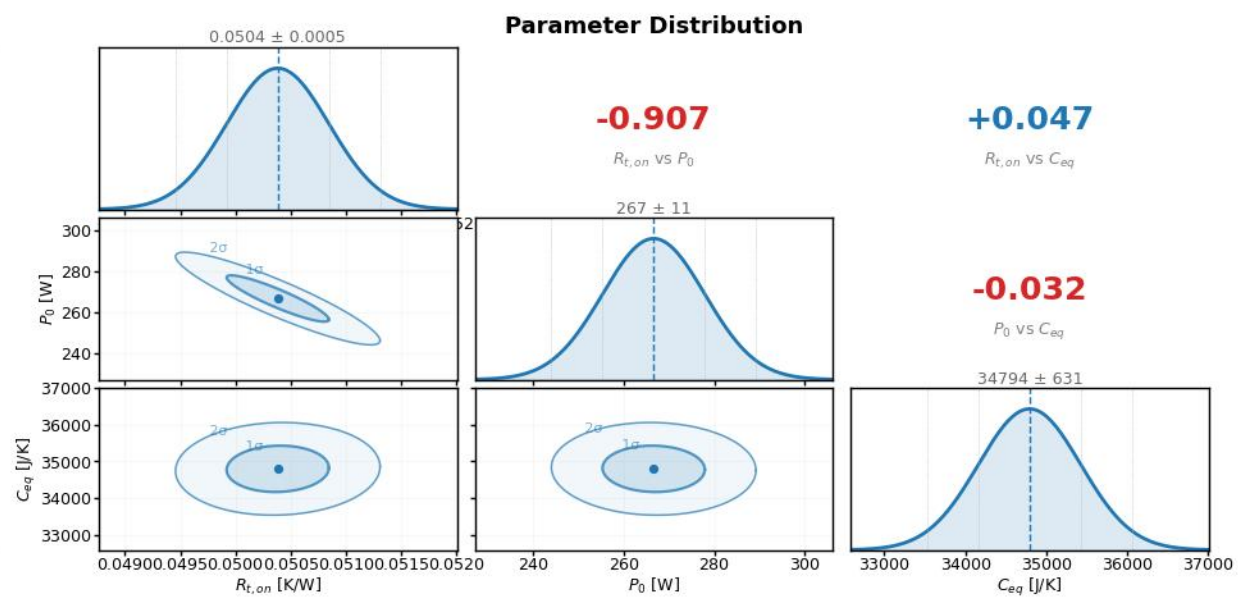
T5 — Machine Study

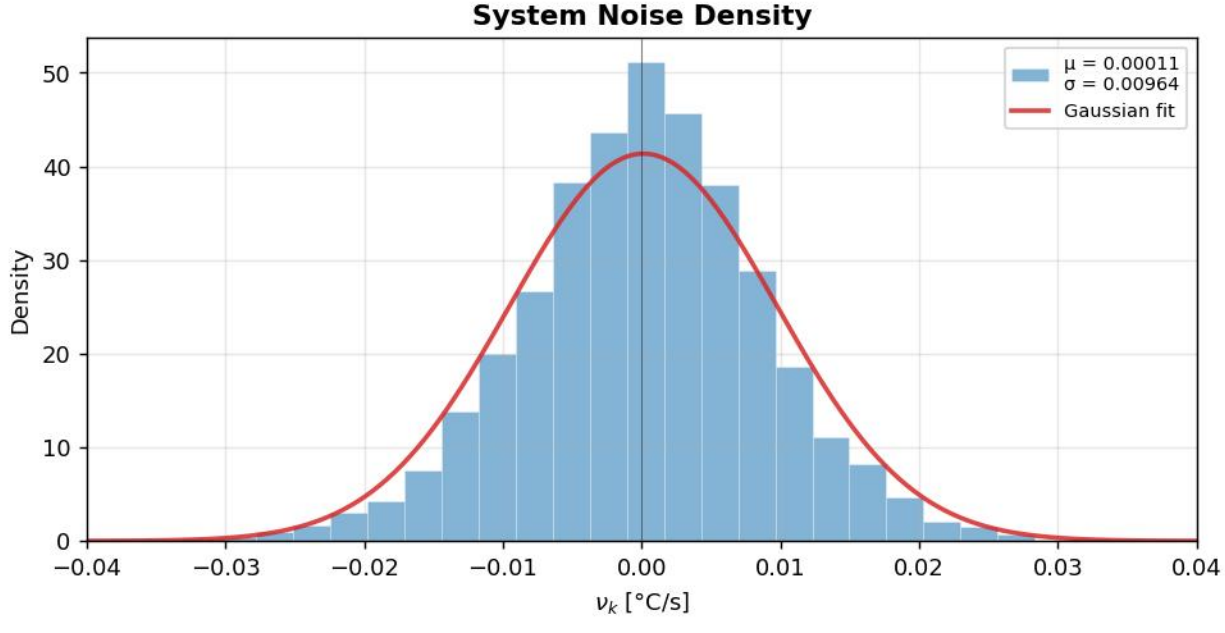












Model Parameters

Fitted parameters

$R_{t,on} = 0.05038 \pm 0.00046 \;\; [°C/W]$

$R_{t,off} = 0.10602 \pm 0.00355 \;\; [°C/W]$

$P_0 = 266.6 \pm 11.3 \;\; [W]$

$C_{eq} = 34794 \pm 631 \;\; [J/°C]$

Derived quantities

$\tau_{on} = R_{t,on} \times C_{eq} = 29.2 \pm 0.6$ min

$I_{cont} = 1.165 \pm 0.003 \;\; \times I_e = 115.9 \pm 0.3$ A

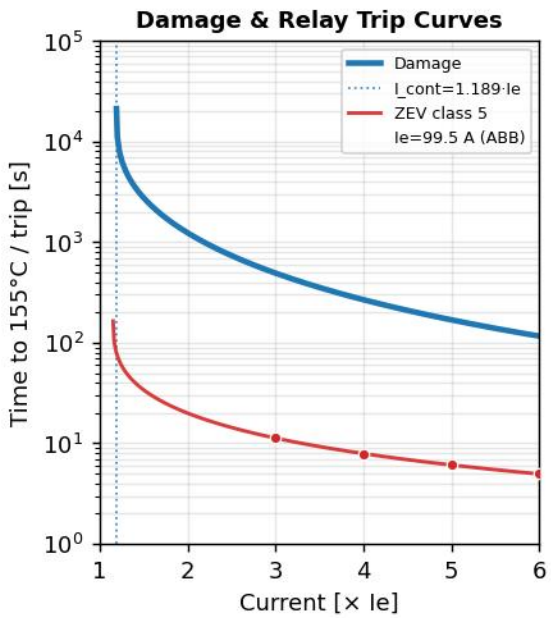
Fit quality

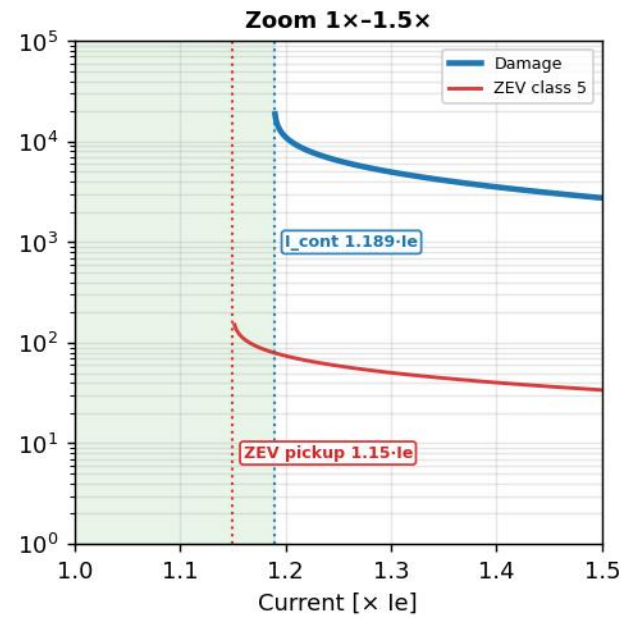
$RMSE = 3.777°C \qquad R^2 = 0.9799 \qquad \sigma(T_w) = 26.67°C$

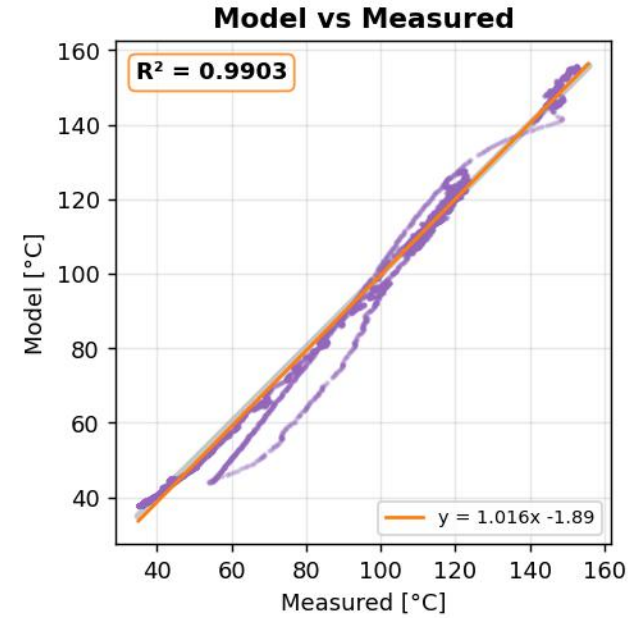
$\sigma_w = 0.00964 \;\; \pm \sim 1\% \;\; [°C/s] \qquad corr(\nu, I) = +0.041$

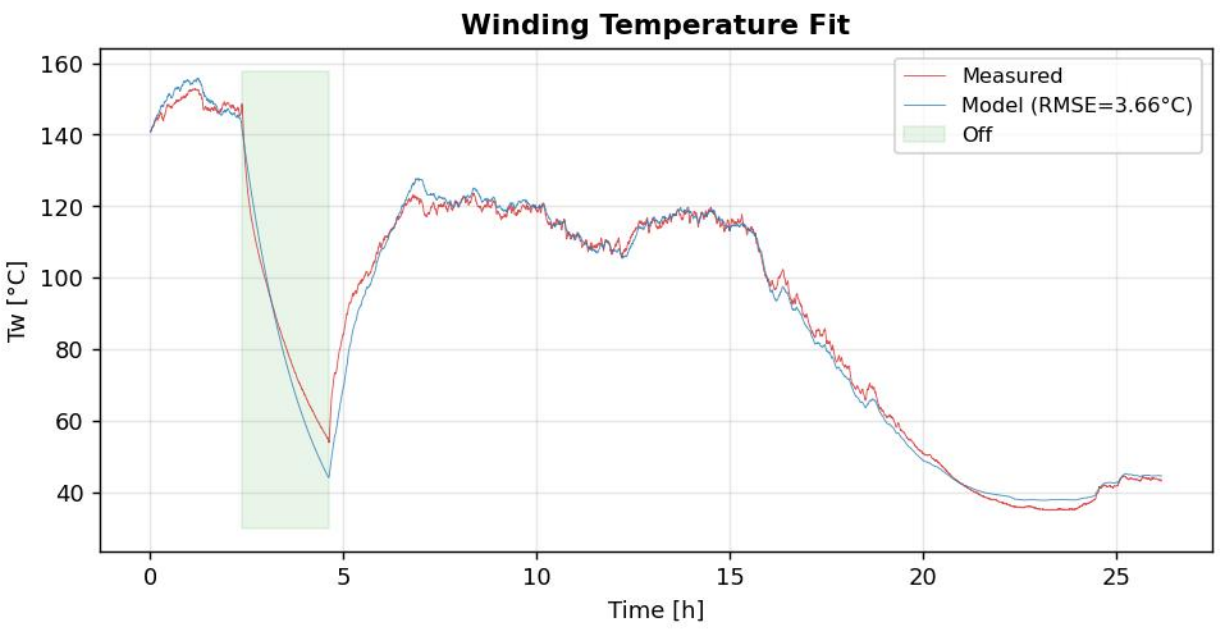
$N_{eff} = 5039 \qquad N = 5039$

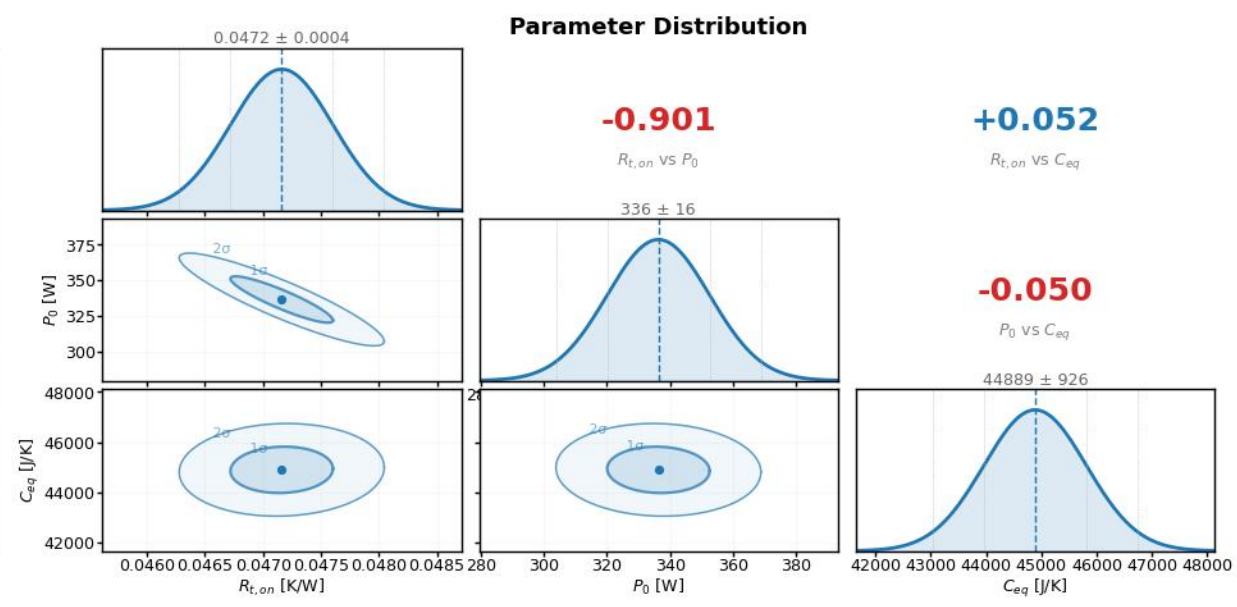
T6 — Machine Study

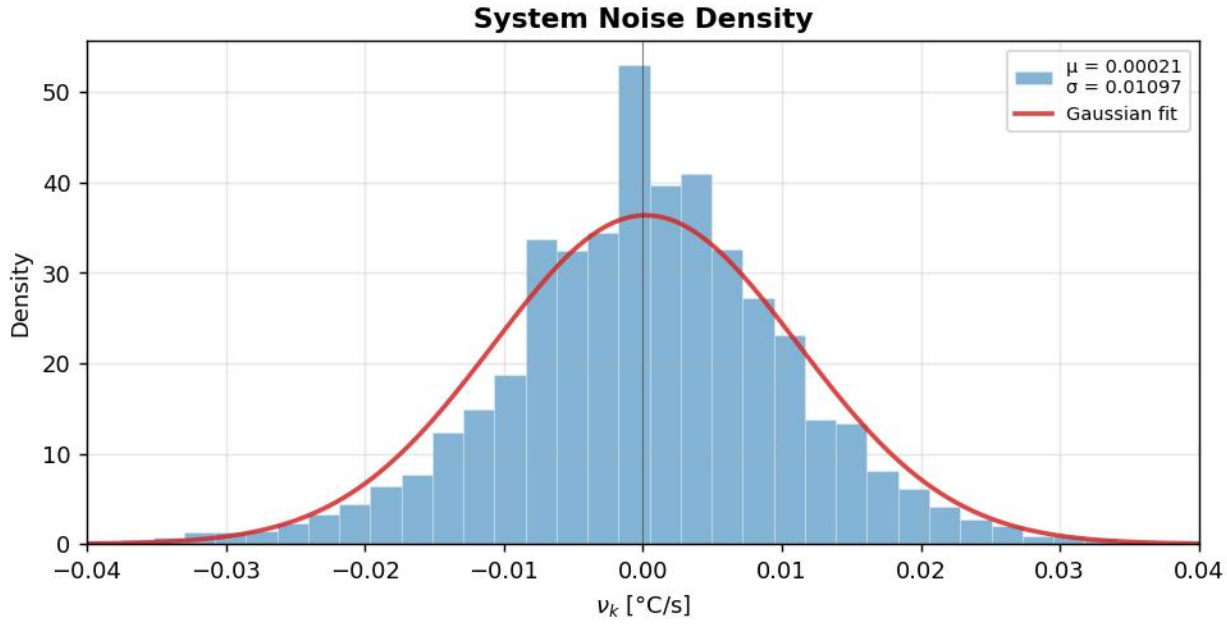












Model Parameters

Fitted parameters

$R_{t,on} = 0.04716 \pm 0.00044 \;\; [°C/W]$

$R_{t,off} = 0.12778 \pm 0.00563 \;\; [°C/W]$

$P_0 = 336.3 \pm 16.3 \;\; [W]$

$C_{eq} = 44889 \pm 926 \;\; [J/°C]$

Derived quantities

$\tau_{on} = R_{t,on} \times C_{eq} = 35.3 \pm 0.8$ min

$I_{cont} = 1.189 \pm 0.003 \;\; \times I_e = 118.3 \pm 0.3$ A

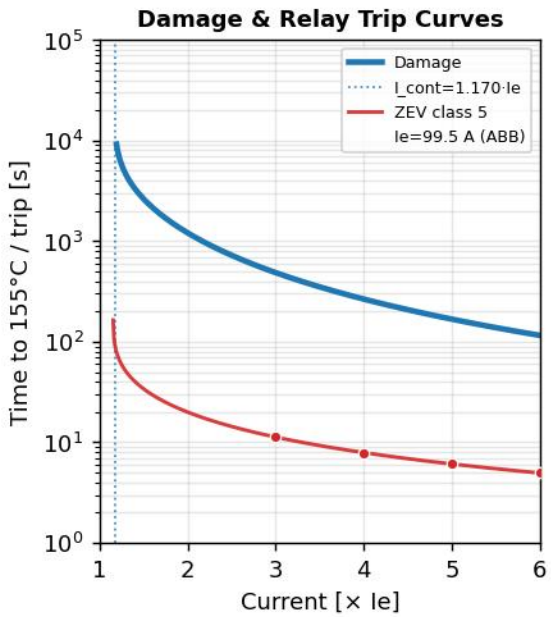
Fit quality

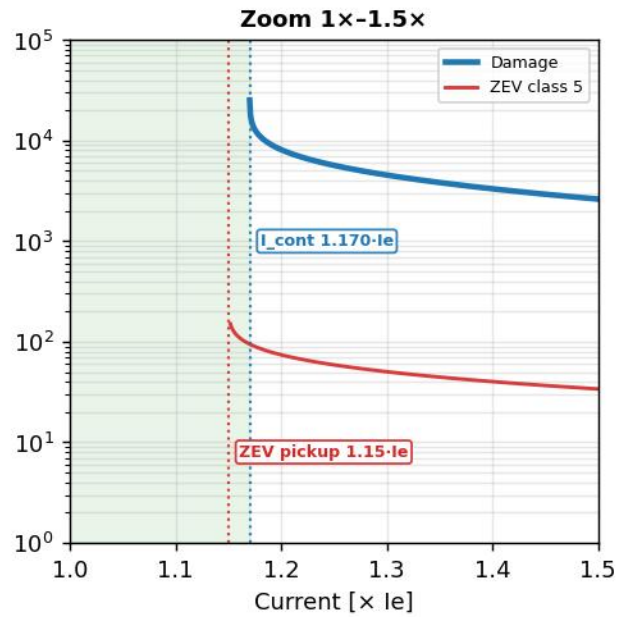
$RMSE = 3.663°C \qquad R^2 = 0.9895 \qquad \sigma(T_w) = 35.72°C$

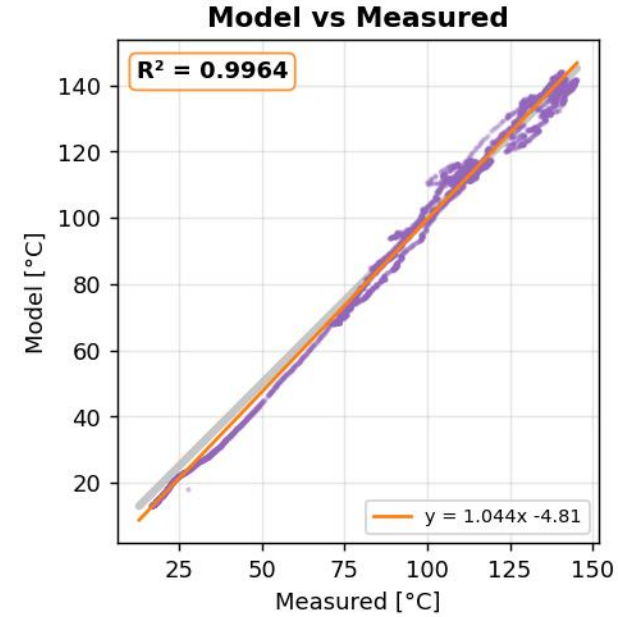
$\sigma_w = 0.01097 \;\; \pm \sim 1\% \;\; [°C/s] \qquad corr(\nu, I) = +0.055$

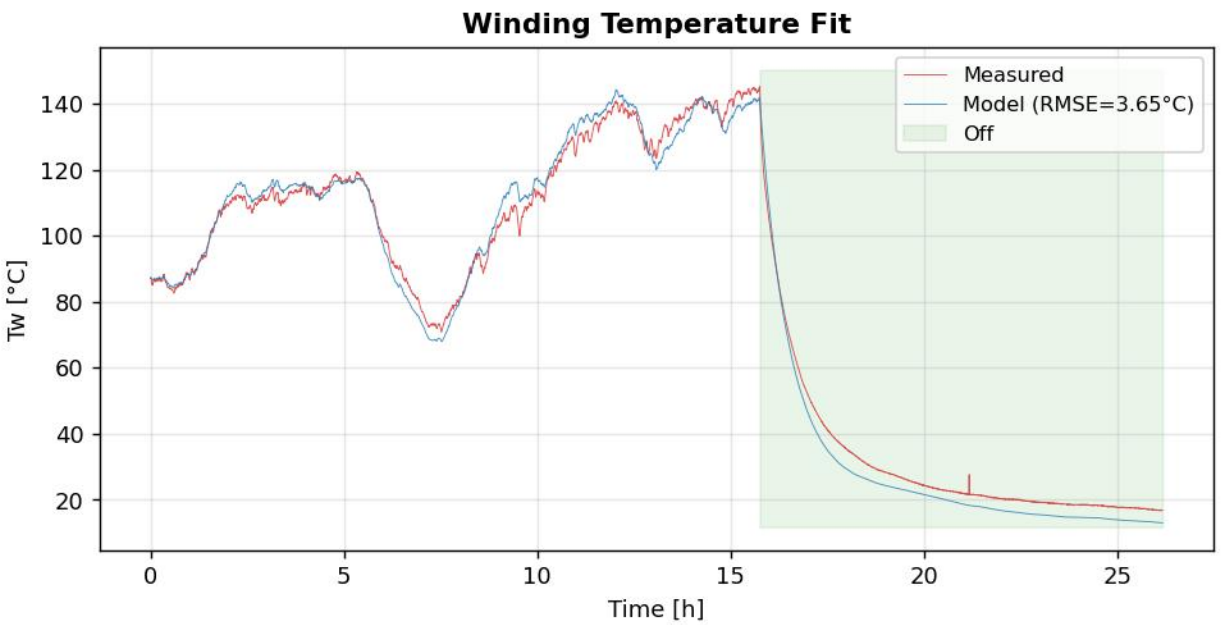
$N_{eff} = 5062 \qquad N = 5062$

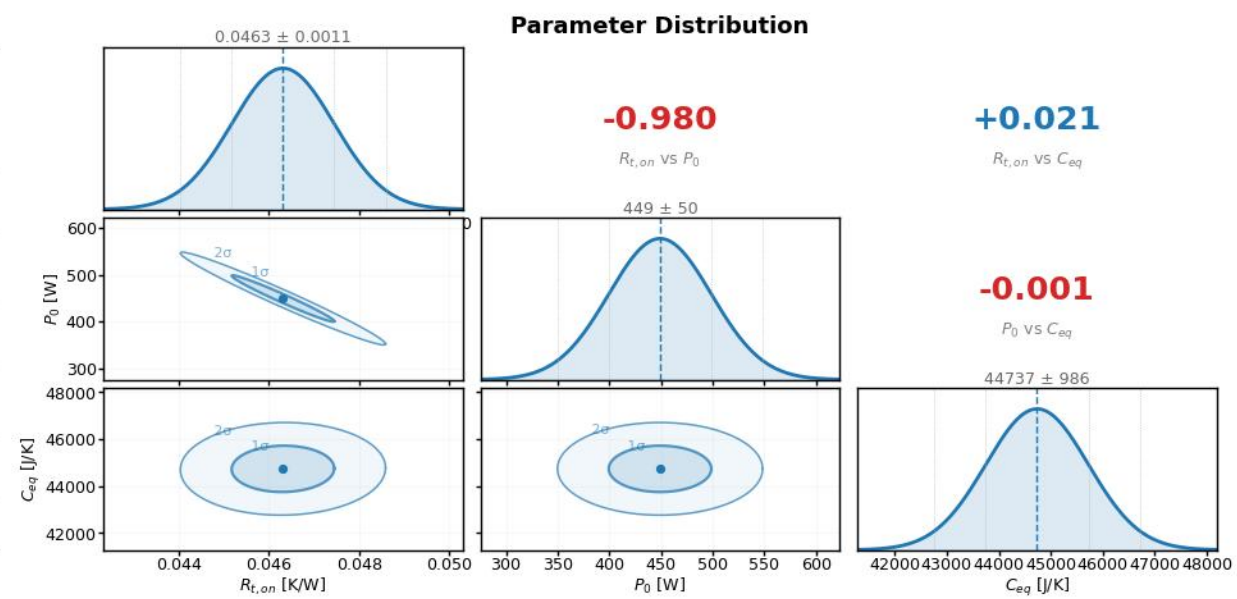
T7 — Machine Study

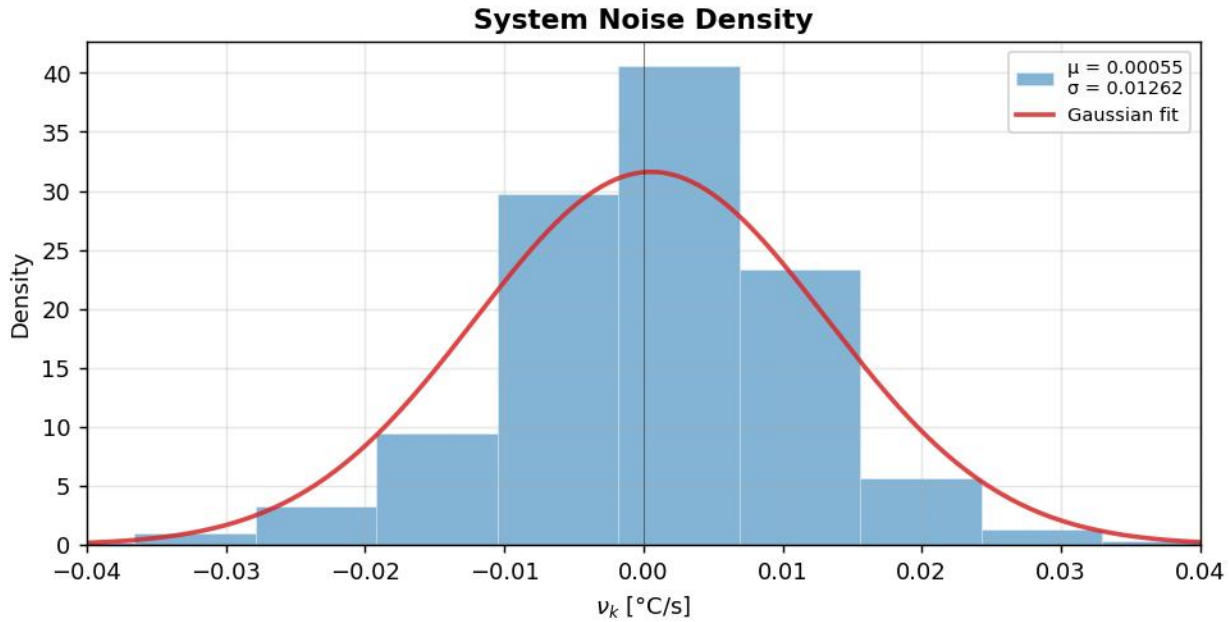












Model Parameters

Fitted parameters

$R_{t,on} = 0.04631 \pm 0.00114 \;\; [°C/W]$

$R_{t,off} = 0.06009 \pm 0.00233 \;\; [°C/W]$

$P_0 = 449.1 \pm 49.7 \;\; [W]$

$C_{eq} = 44737 \pm 986 \;\; [J/°C]$

Derived quantities

$\tau_{on} = R_{t,on} \times C_{eq} = 34.5 \pm 1.2$ min

$I_{cont} = 1.170 \pm 0.004 \;\times I_e = 116.4 \pm 0.4$ A

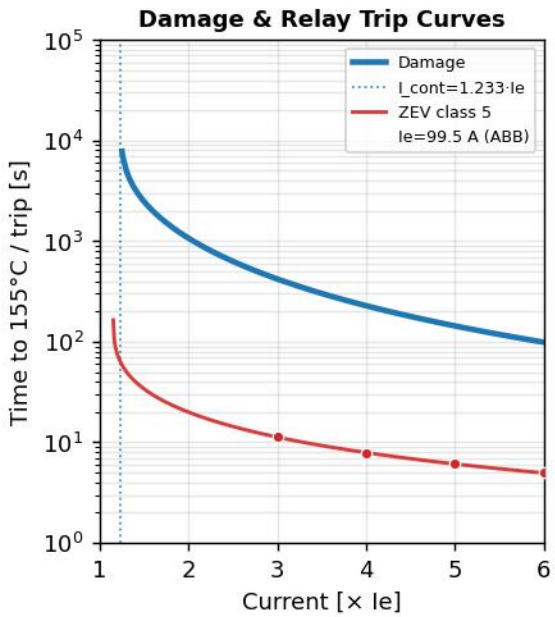
Fit quality

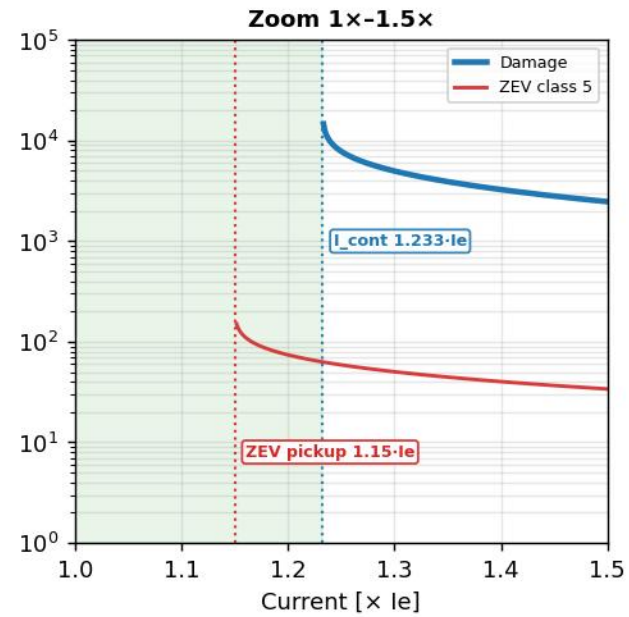
$RMSE = 3.654°C \qquad R^2 = 0.9933 \qquad \sigma(T_w) = 44.56°C$

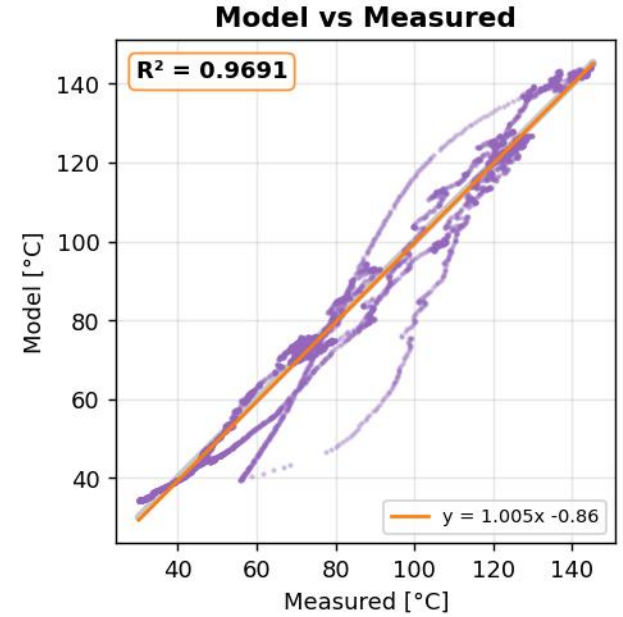
$\sigma_w = 0.01262 \;\pm\sim 1\% \;\; [°C/s] \qquad corr(\nu, I) = +0.047$

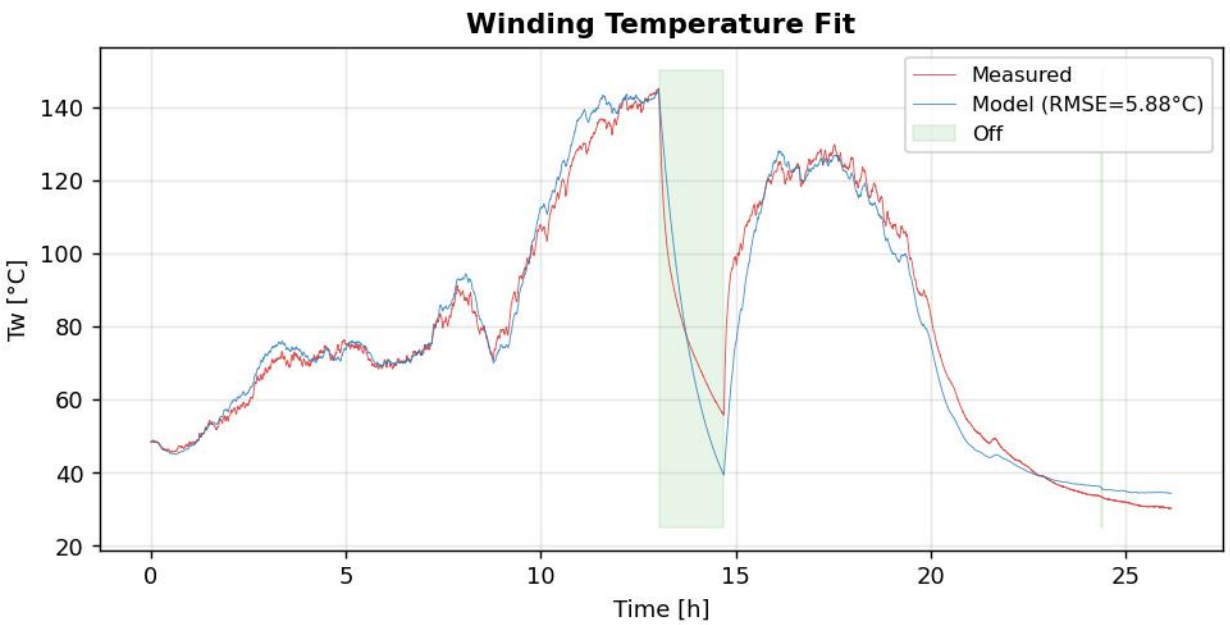
$N_{eff} = 5154 \qquad N = 5154$

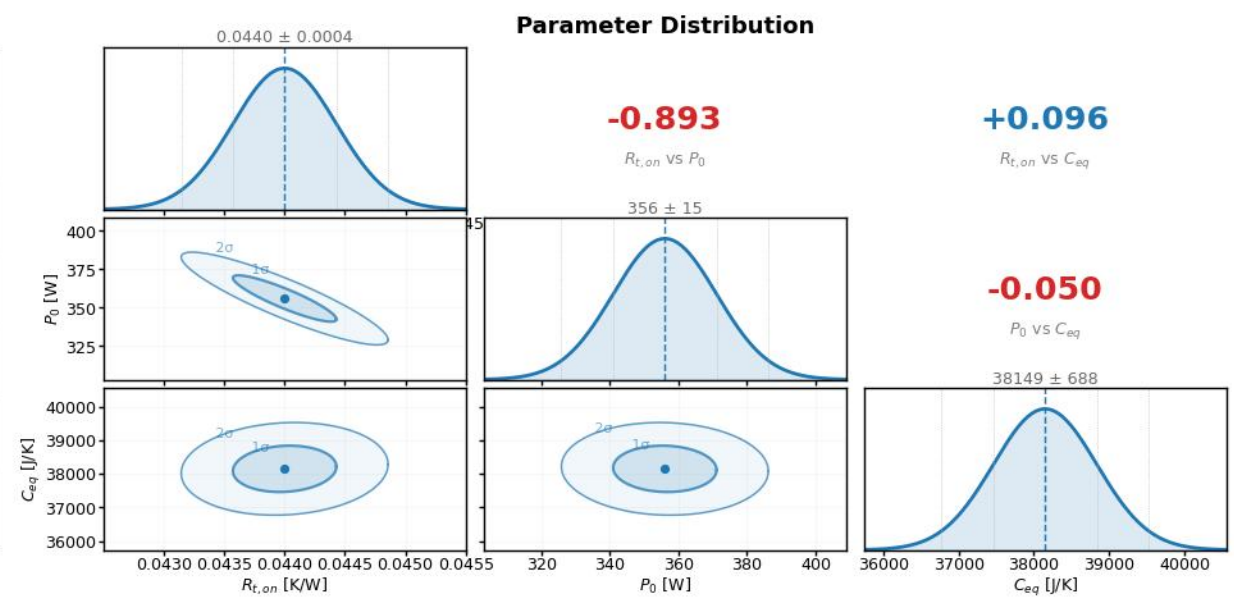
T8 — Machine Study

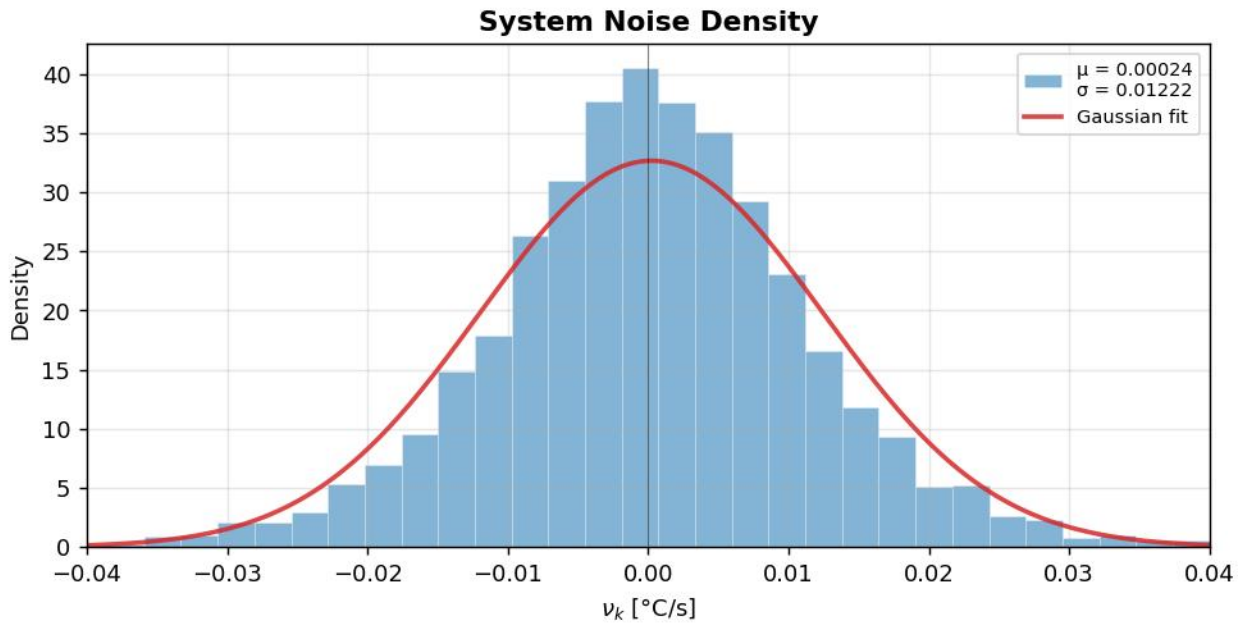












Model Parameters

Fitted parameters

$R_{t,on} = 0.04400 \pm 0.00043 \quad [°C/W]$

$R_{t,off} = 0.08719 \pm 0.00353 \quad [°C/W]$

$P_0 = 356.0 \pm 15.1 \quad [W]$

$C_{eq} = 38149 \pm 688 \quad [J/°C]$

Derived quantities

$\tau_{on} = R_{t,on} \times C_{eq} = 28.0 \pm 0.6$ min

$I_{cont} = 1.233 \pm 0.004 \times I_e = 122.6 \pm 0.4$ A

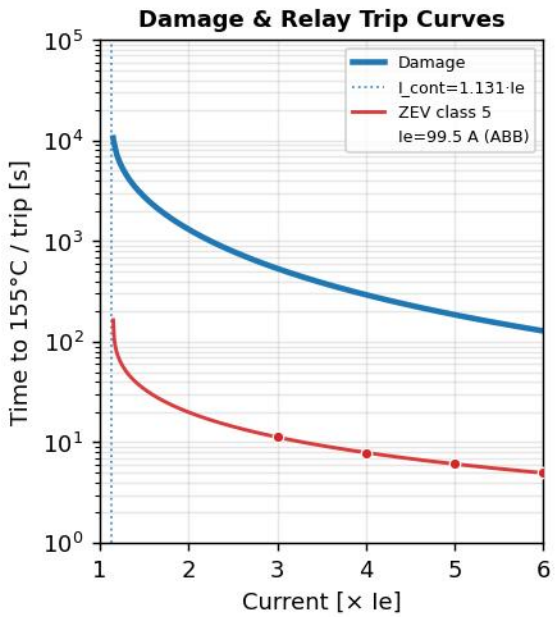
Fit quality

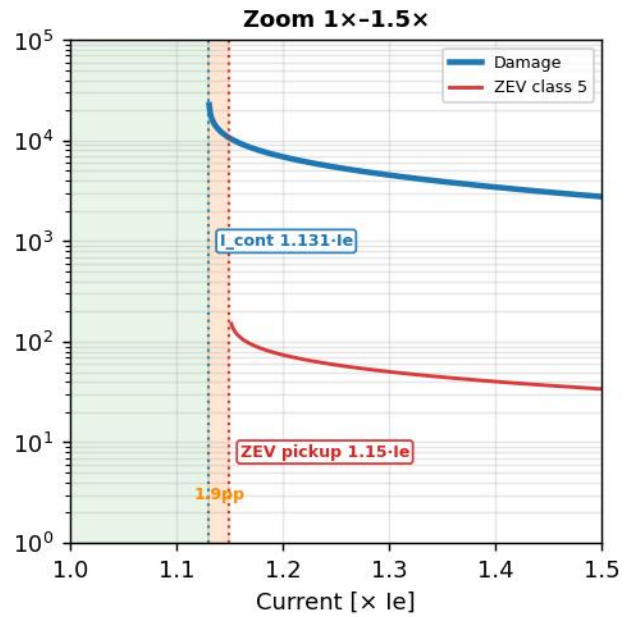
$RMSE = 5.884°C \quad R^2 = 0.9676 \quad \sigma(T_w) = 32.68°C$

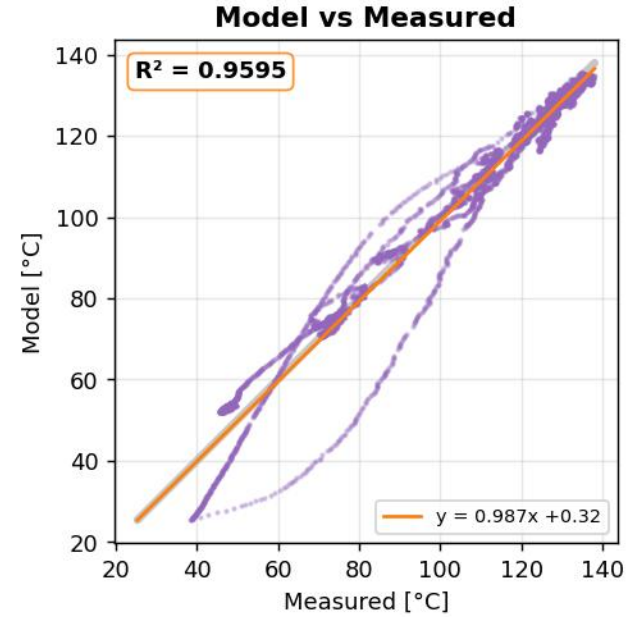
$\sigma_w = 0.01222 \pm \sim 1\% \quad [°C/s] \quad corr(\nu, I) = +0.046$

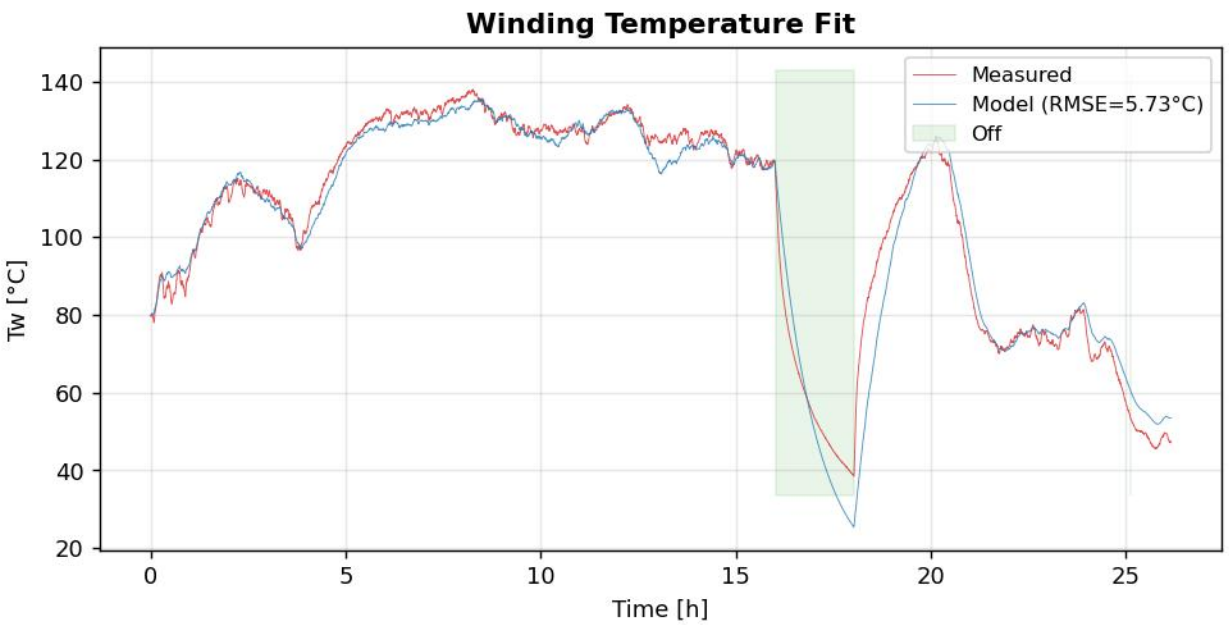
$N_{eff} = 4993 \quad N = 4993$

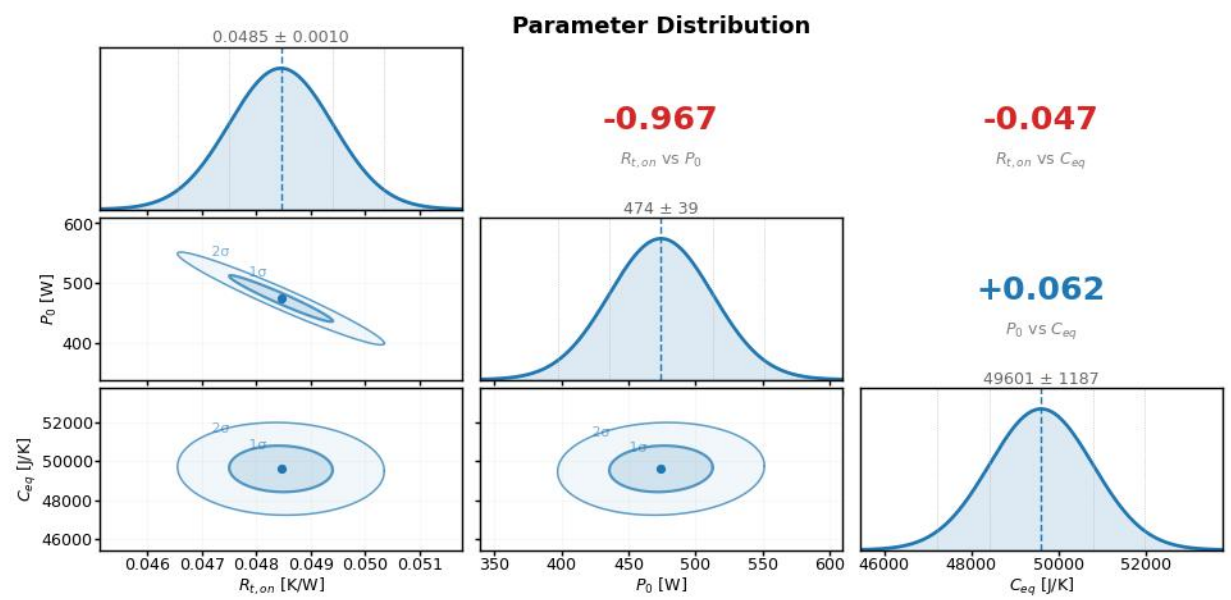
T9 — Machine Study

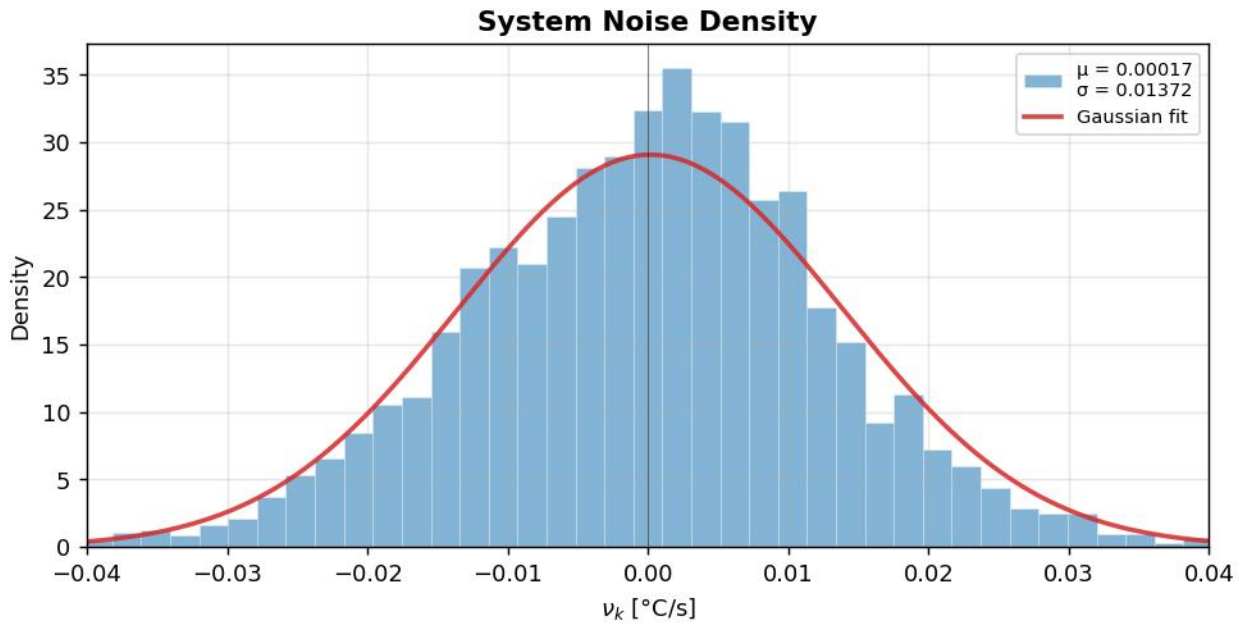












Model Parameters

Fitted parameters

$R_{t,on} = 0.04845 \pm 0.00095 \quad [°C/W]$

$R_{t,off} = 0.06960 \pm 0.00374 \quad [°C/W]$

$P_0 = 474.3 \pm 38.6 \quad [W]$

$C_{eq} = 49601 \pm 1187 \quad [J/°C]$

Derived quantities

$\tau_{on} = R_{t,on} \times C_{eq} = 40.1 \pm 1.2$ min

$I_{cont} = 1.131 \pm 0.004 \times I_e = 112.5 \pm 0.4$ A

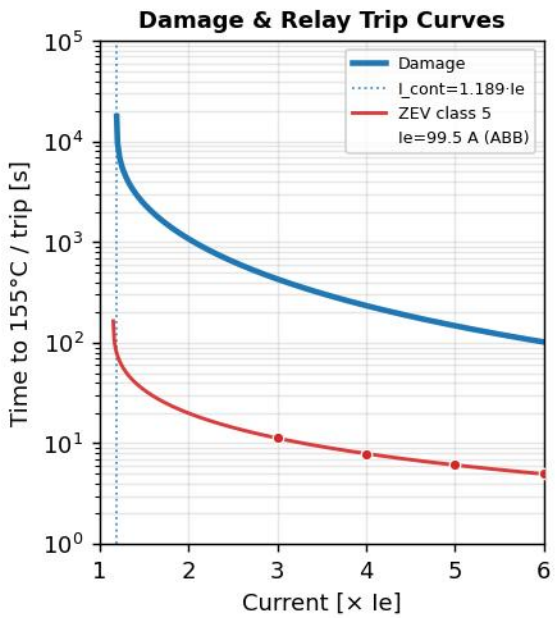
Fit quality

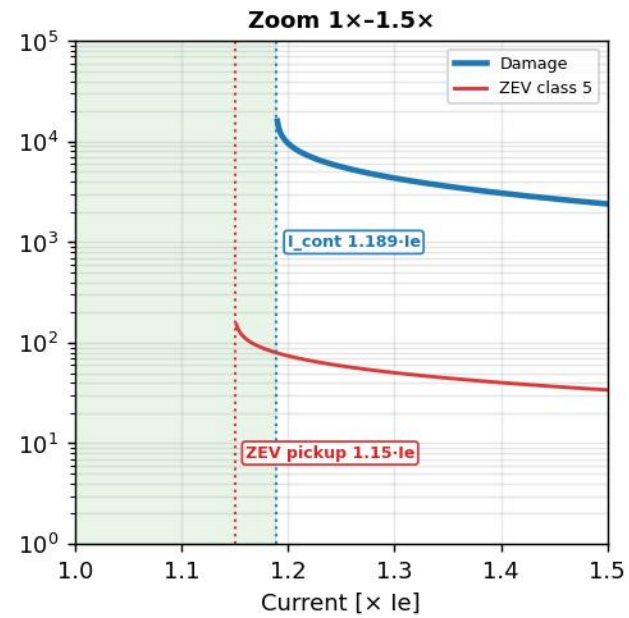
$RMSE = 5.733°C \quad R^2 = 0.9574 \quad \sigma(T_w) = 27.79°C$

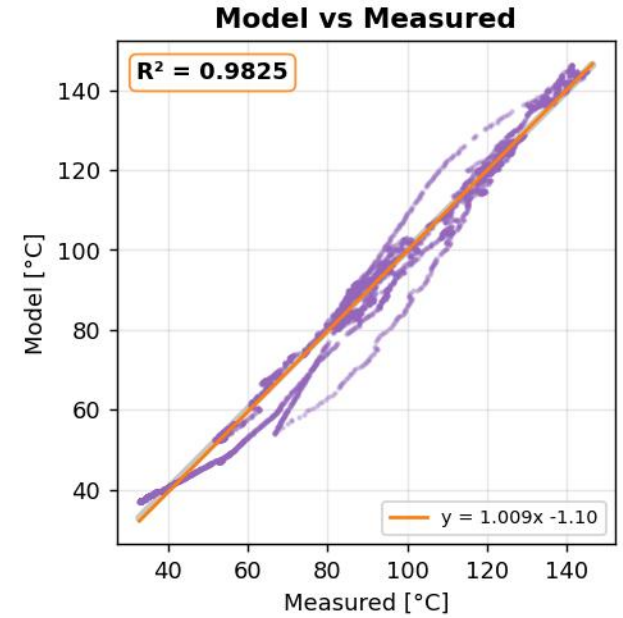
$\sigma_w = 0.01372 \pm \sim 1\% \quad [°C/s] \quad corr(\nu, I) = +0.052$

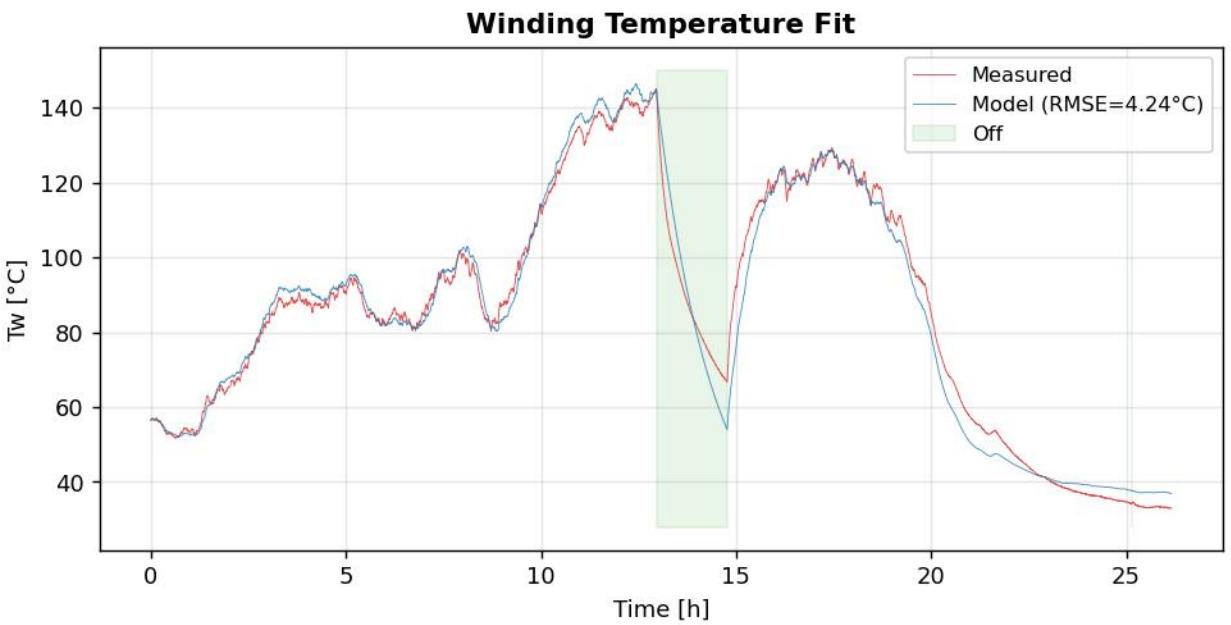
$N_{eff} = 4971 \quad N = 4971$

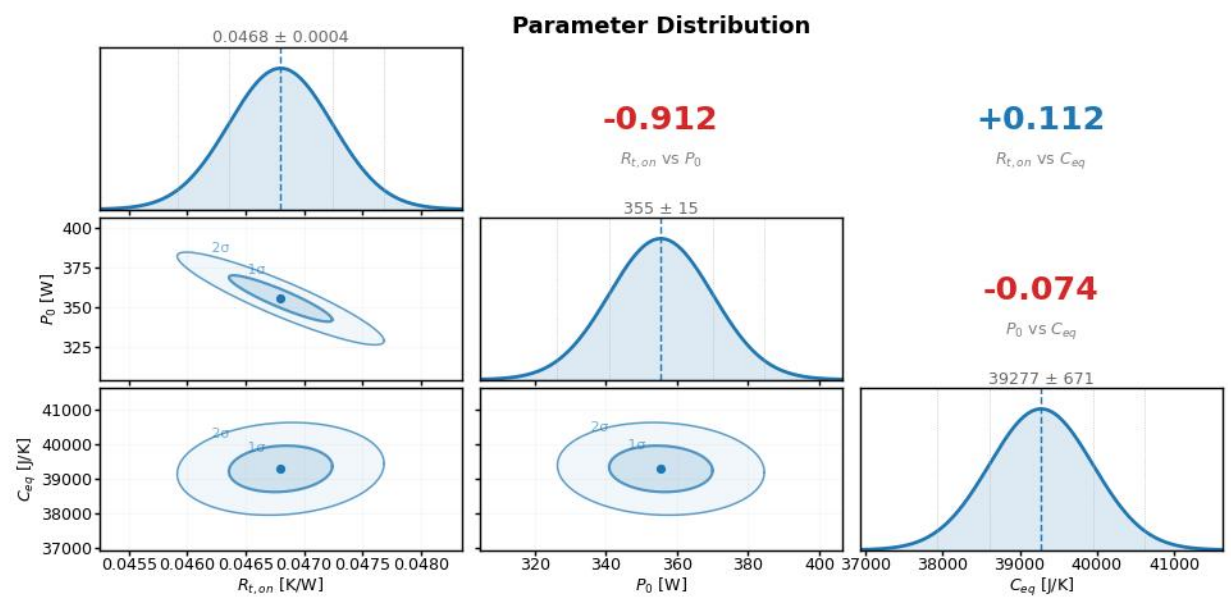
T10 — Machine Study

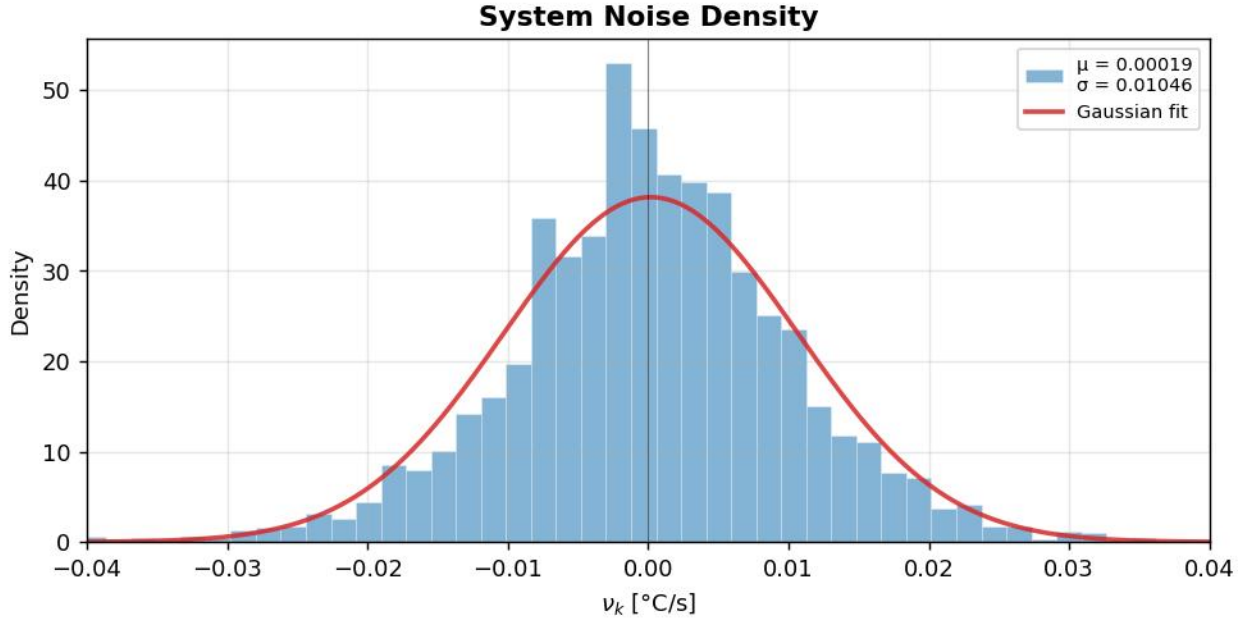












Model Parameters

Fitted parameters

$R_{t,on} = 0.04680 \pm 0.00044 \quad [°C/W]$

$R_{t,off} = 0.12889 \pm 0.00553 \quad [°C/W]$

$P_0 = 355.5 \pm 14.5 \quad [W]$

$C_{eq} = 39277 \pm 671 \quad [J/°C]$

Derived quantities

$\tau_{on} = R_{t,on} \times C_{eq} = 30.6 \pm 0.6$ min

$I_{cont} = 1.189 \pm 0.003 \times I_e = 118.3 \pm 0.3$ A

Fit quality

$RMSE = 4.245°C \quad R^2 = 0.9817 \quad \sigma(T_w) = 31.37°C$

$\sigma_w = 0.01046 \pm \sim 1\% \quad [°C/s] \quad corr(\nu, I) = +0.049$

$N_{eff} = 4945 \quad N = 4945$

Model Equations Summary

1. Thermal SDE

$$C\frac{dT}{dt} = I^2 R_{dc}(T) + P_0 - \frac{T - T_{amb}}{R_t} + \sigma_w \frac{dB}{dt}$$

$R_t = R_{t,on}\mathbf{1}_{\{on\}} + R_{t,off}\mathbf{1}_{\{off\}}, \quad P_0 \to P_0\mathbf{1}_{\{on\}}, \quad dB/dt$: white noise

2. Thermal ODE

$$\frac{dT}{dt} = a\,T + b(t) \quad a = \frac{I^2 R_{dc,20}\alpha - 1/R_t}{C} \quad b = \frac{I^2 R_{dc,20}(1-\alpha T_{ref}) + P_0 + T_{amb}/R_t}{C}$$

$R_{dc}(T) = R_{dc,20}[1 + \alpha(T - T_{ref})],\ T_{ref} = 20°C,\ \alpha = 0.00393/°C$

Steady state at T_{lim}: $I_{cont}^2 R_{dc}(T_{lim}) + P_0 = (T_{lim} - T_{amb})/R_{t,on}$

$\Rightarrow\ I_{cont} = \sqrt{[(T_{lim} - T_{amb})/R_{t,on} - P_0]\,/\,R_{dc}(T_{lim})}$ $\quad (T_{lim} = 155°C,$ class F)

3. Trapezoidal (exponential) update

$$T_{k+1} = e^{a\Delta t}\left(T_k + \frac{\Delta t}{2}b_k\right) + \frac{\Delta t}{2}b_{k+1}$$

4. Innovation (whitened residual)

$$e_k = T_{meas,k} - T_{model,k} \quad \nu_k = \frac{e_{k+1} - e^{a\Delta t}e_k}{\Gamma_k},\ \Gamma_k = \frac{e^{a\Delta t} - 1}{a} \quad \sigma_w = \sqrt{\frac{1}{N_{eff}}\sum_{k=1}^{N_{eff}}(\nu_k - \bar{\nu})^2}$$

5. Relay and Machine Trip Time (t_A)

$$t_A^{ZEV} = \frac{CLASS}{5} \times 23.3\ln\left(\frac{I/I_e}{I/I_e - 1.15}\right), \quad t_A^{mach} = \frac{1}{a}\ln\left(\frac{T_{lim} - T_{ss}}{T_0 - T_{ss}}\right),\ T_{ss} = -b/a$$

6. Error statistics (first − order propagation)

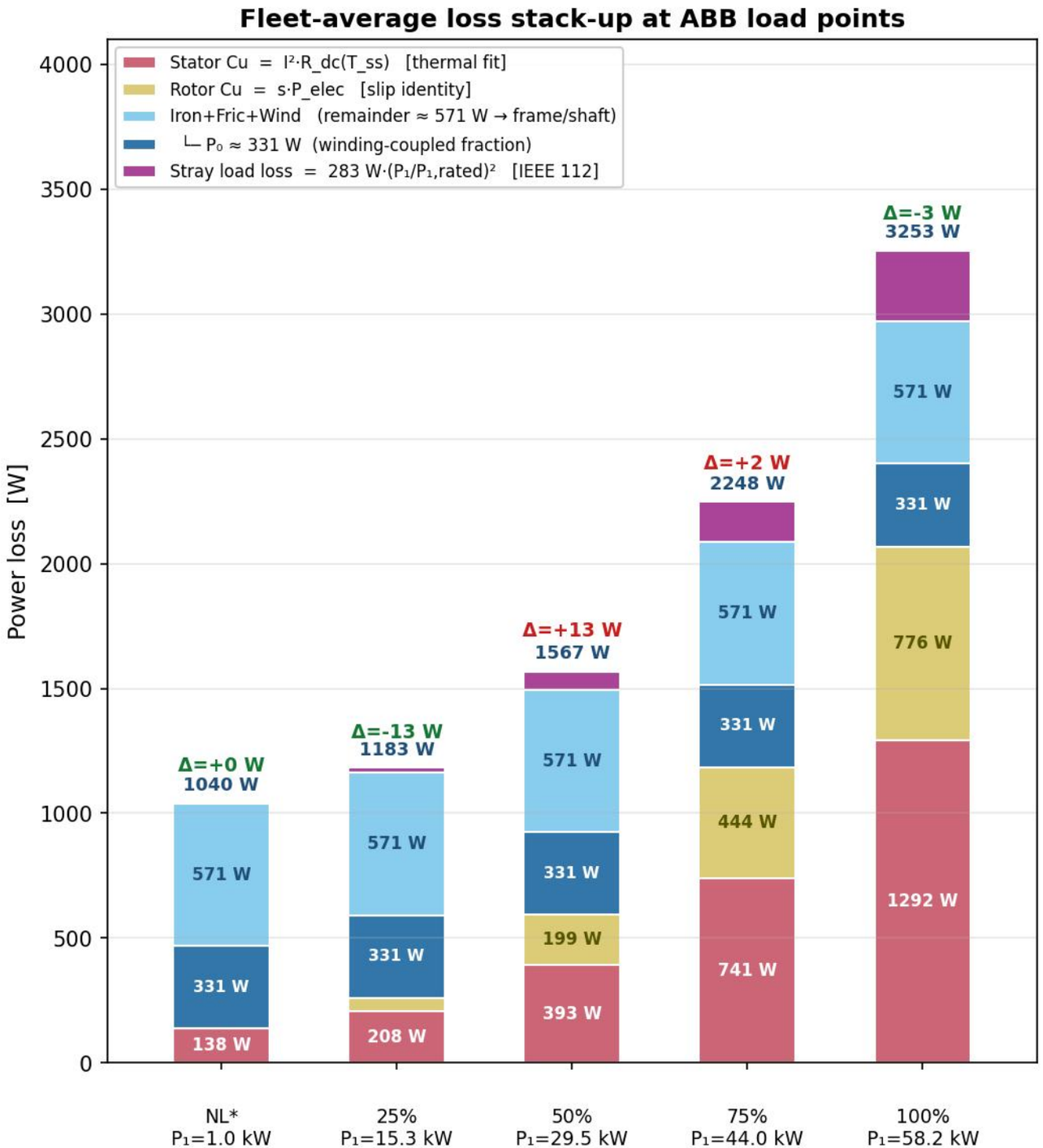
$$\sigma_f^2 = J\Sigma J^\top = \sum_{i,j}\frac{\partial f}{\partial x_i}\frac{\partial f}{\partial x_j}\Sigma_{ij} \qquad \Sigma = \mathrm{diag}(\sigma)\cdot\mathrm{corr}\cdot\mathrm{diag}(\sigma)$$

$$J_{\tau_{on}} = [C_{eq},\ 0,\ 0,\ R_{t,on}] \qquad J_{I_{cont}} = [\partial I_{cont}/\partial R_{t,on},\ 0,\ \partial I_{cont}/\partial P_0,\ 0]$$

$$R^2 = 1 - \frac{RMSE^2}{\sigma^2(T_w)} \qquad N_{eff} = \sum_{k=1}^{N}\mathbf{1}(w_k = 1)$$

$w_k = 0$ if $\Delta t_{k-1} > 300$ s (data gap), $\quad w_k = 1$ otherwise

Efficiency Budget: Fleet-Average Loss Stack-Up vs ABB Datasheet



Loss decomposition equations

1. Stator Cu (1C1R thermal fit):

$$P_{Cu,s} = I^2 R_{dc}(T_{ss}), \quad R_{dc}(T) = R_{dc,20}[1+\alpha(T-20)]$$

2. Rotor Cu (slip identity, no temp correction):

$$P_{rCu} = s \cdot P_{elec} \quad \text{(energy conservation at air gap)}$$

2a. Slip linearity (linear s–P_1):

$$s(P_1) = s_{rated} \cdot P_1/P_{1,rated}, \quad s_{rated} = 0.01333, \quad P_{1,rated} = 58.25\,\text{kW}$$

3. Iron + Fric + Wind (ABB no-load test):

$$P_{ifw} = P_{NL} - 3\,I_{NL}^2 R_1(T_{w,NL})$$

$$(T_{w,NL} - T_{amb})/R_{t,on} = 3 I_{NL}^2 R_1(T_{w,NL}) + P_0$$

$$\Rightarrow T_{w,NL} = 48.3\,°C, \quad P_{ifw} = 1040 - 138 = \mathbf{902\ W}$$

note: $P_0 \subset P_{ifw}$ (winding-coupled fraction shown in dark blue)

4. Stray Load Loss (IEEE 112 §5.3, ABB "PLL from residual"):

$$P_{stray}(P_1) = P_{stray,rated} \cdot (P_1/P_{1,rated})^2, \quad P_{stray,rated} = 283\,\text{W}$$

(equivalent to 0.486% of rated input; fitted from 3-term residual)

Prediction: $\mathbf{P_{pred} = P_{Cu,s} + P_{rCu} + P_{ifw} + P_{stray}}$

Result summary

Load	Pred	ABB (P_1–P_2)	Δ
NL*	1040 W	P_NL = 1.04 kW	+0 W
25%	1183 W	15.30–14.13=1.17 kW	-13 W
50%	1567 W	29.52–27.94=1.58 kW	+13 W
75%	2248 W	44.04–41.79=2.25 kW	+2 W
100%	3253 W	58.25–55.00=3.25 kW	-3 W

* NL row: anchor (closes by construction)
P_NL from No-Load test; P_1, P_2 from
Partial Load Points (ABB Type Test Report)